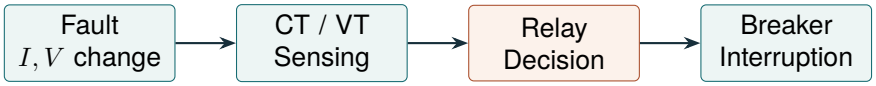
CET / SEMESTER 5 / PCEOT503

POWER SYSTEMS

Module 3

Fault analysis, protective relays and circuit breakers



Learn the calculation and the physical response together

Per-unit quantities; symmetrical components; sequence networks; every supplied class fault problem; protection principles; induction, static and numerical relays; and a complete syllabus supplement on arc extinction, SF_6 and vacuum circuit breakers.

Comprehensive study notes covering per-unit systems, symmetrical components, fault analysis, protective relaying, and circuit breakers according to the 2024-scheme syllabus.

Prepared for Sudheesh / 8 October 2026

Contents

1 Study map and notation

1.1 Everything required by the supplied Module 3 syllabus

Module 3 has 11 contact hours and combines analysis with protection. The analysis predicts currents and voltages during faults; protection uses measurements to decide which breakers should interrupt those currents.

Syllabus item	What this note develops
Per-unit quantities: single and three phase	Base definitions, change of base, transformer referral, complete class examples.
Symmetrical components and sequence networks	Forward/inverse transformations, phasor sets, component models, grounding and transformer zero-sequence paths.
Fault types and shunt-fault calculations	Three-phase, LG, LL and LLG boundary-condition derivations; solid and finite-impedance faults; every supplied class numerical. Series/open-conductor fault calculations are outside scope.
Need for protection; primary and backup	Selectivity, reliability, stability, sensitivity, speed, zones, coordination and breaker failure.
Induction-type electromechanical relay	Construction, phase-displaced flux, torque, disc/cup structures, pickup and time settings.
Static OC and instantaneous OC	Labelled functional block diagrams, comparators, timing and trip chain.
Microprocessor OC; numerical relay	Hardware diagram, flowchart, sampling, digital estimation, timing, functions and limitations.
OC, directional, distance, differential	Measured quantities, decision principles, applications, characteristic sketches and examples.
Circuit breaker; arc; extinction principles/methods	Contact separation, current zero, recovery voltage, thermal/dielectric recovery, high/low-resistance methods.
SF_6 and VCB	Construction, interruption sequence, advantages/limitations and comparison.

Key analytical pitfalls to avoid

In unsymmetrical fault analysis, double-line-to-ground (LLG) calculations require strict adherence to sequence current sign conventions to ensure zero fault current in the unfaulted phase ($I_a = 0$). Section **??** works through the complete boundary conditions and verification checks.

1.2 Notation and assumptions used throughout

- $a = e^{\mathrm{j}120^\circ}$ is the *sequence rotation operator*, not a transformer turns ratio. Use $t = N_H/N_L$ for the latter.
- Sequence ordering is $(0, 1, 2)$: zero, positive, negative. V_1 means the phase-a positive-sequence voltage; the other two phases follow by rotation. I_k has the same convention.
- E is the pre-fault positive-sequence Thevenin phase voltage at the fault bus. Z_1, Z_2, Z_0 are the driving-point sequence impedances seen there on a common base. They are not automatically the reactance of just one generator.
- Fault currents are directed from the network *into the fault*. We analyse an a-ground fault, a b-c fault and a b-c-ground fault. Relabelling phases changes phasors, not the method.
- Unless stated otherwise: balanced pre-fault network, $E = 1\angle 0^\circ$ p.u., no pre-fault load-current contribution, balanced sequence-decoupled equipment, and RMS phasors. Reactance-only examples neglect resistance. These are textbook calculation assumptions.

Check your understanding

Why is a fault calculation not enough to design protection?
Answer: It predicts electrical stresses under a stated network model. Relay decisions also need operating settings, selectivity, instrument-transformer performance, breaker clearing capability and contingency/backup analysis.

2 Per-unit quantities without hidden factors

Class-slide anchor: Per-unit part pp. 1–16.

2.1 The idea: normalise quantities to a consistent base

$$x_{\text{pu}} = \frac{x_{\text{actual}}}{x_B}, \qquad x_{\text{actual}} = x_{\text{pu}} x_B, \qquad x_{\%} = 100 x_{\text{pu}}.$$

A reactance of 20% is 0.2 p.u. on the base for which it was specified. "Per unit" has no unit, but it is meaningless without its base. 0.2 p.u. on 10 MVA and 0.2 p.u. on 100 MVA are not the same physical impedance if the voltage bases match.

A network has machines, transformers and lines at different voltage levels. Ohmic impedances must be referred across every transformer before direct combination. Per-unit values avoid repeated referral *after* we choose consistent bases. Apparatus ratings and typical p.u. reactances also become easier to compare.

2.2 Single-phase base quantities

Choose base apparent power S_B and RMS voltage V_B independently. Then

$$S_B = V_B I_B, \qquad I_B = \frac{S_B}{V_B}, \qquad Z_B = \frac{V_B}{I_B} = \frac{V_B^2}{S_B}, \qquad Y_B = \frac{1}{Z_B}.$$

With S_B in kVA and V_B in kV:

$$I_B = \frac{S_B(\text{kVA})}{V_B(\text{kV})}\ \text{A}, \qquad Z_B = \frac{1000 V_B^2(\text{kV})}{S_B(\text{kVA})}\ \Omega.$$

A genuine single-phase circuit uses its own terminal voltage and total single-phase VA. A one-phase equivalent of a balanced three-phase system uses phase voltage and one-third of total VA. These are two applications of the same formula; single phase does not inherently imply a line-to-neutral connection.

2.3 Three-phase bases and the cancellation of square root three

Use total three-phase S_B and line-to-line voltage magnitude $V_{B,LL}$. The phase equivalent has

$$S_{B,\text{phase}} = S_B/3, \qquad V_{B,\text{phase}} = V_{B,LL}/\sqrt{3}.$$

Therefore

$$\boxed{I_B = \frac{S_B}{\sqrt{3} V_{B,LL}}}, \qquad \boxed{Z_B = \frac{(V_{B,LL}/\sqrt{3})^2}{S_B/3} = \frac{V_{B,LL}^2}{S_B}}.$$

Z_B is the *per-phase equivalent impedance base*. Convenient engineering forms are

$$I_B(\text{kA}) = \frac{S_B(\text{MVA})}{\sqrt{3} V_{B,LL}(\text{kV})}, \quad Z_B(\Omega) = \frac{V_{B,LL}^2(\text{kV})}{S_B(\text{MVA})}.$$

Using kVA/kV instead gives current in amperes and requires the factor 1000 in Z_B. Do not insert another $\sqrt{3}$ into the impedance formula. For a physical delta-connected branch impedance, first convert to its star phase equivalent or explicitly define delta-branch bases; the network's standard per-phase Z_B is not a declaration that every winding is physically star connected.

For consistent phase-reference phasors in a balanced system,

$$S_{\text{pu}} = V_{\text{pu}} I_{\text{pu}}^*, \qquad V_{\text{pu}} = Z_{\text{pu}} I_{\text{pu}}.$$

Here S_{pu} is total three-phase complex power divided by total S_B; V_{pu} uses phase voltage divided by its phase base. The phase/line *magnitudes* have the same p.u. value. Their phasor reference angles need attention: V_{ab} is displaced by 30° from V_a in positive sequence.

2.4 Derive change of base rather than memorising its direction

The physical impedance stays the same:

$$Z_{\text{actual}} = Z_{\text{pu,old}} \frac{V_{B,old}^2}{S_{B,old}}.$$

Divide by the new base impedance:

$$\boxed{Z_{\text{pu,new}} = Z_{\text{pu,old}} \frac{S_{B,new}}{S_{B,old}} \left(\frac{V_{B,old}}{V_{B,new}} \right)^2}.$$

Voltage bases must refer to the same physical side when using this formula directly. Increasing base MVA increases p.u. impedance; increasing base voltage decreases it quadratically. A p.u. impedance can be complex; apply the real base multiplier to both R and X.

2.5 Why an ideal transformer disappears from a p.u. diagram

Let $t = N_H/N_L = V_H/V_L$ for a single-phase ideal transformer. Current magnitudes obey $I_L = tI_H$. Equal physical loss and reactive-power effect require

$$Z_H = t^2 Z_L.$$

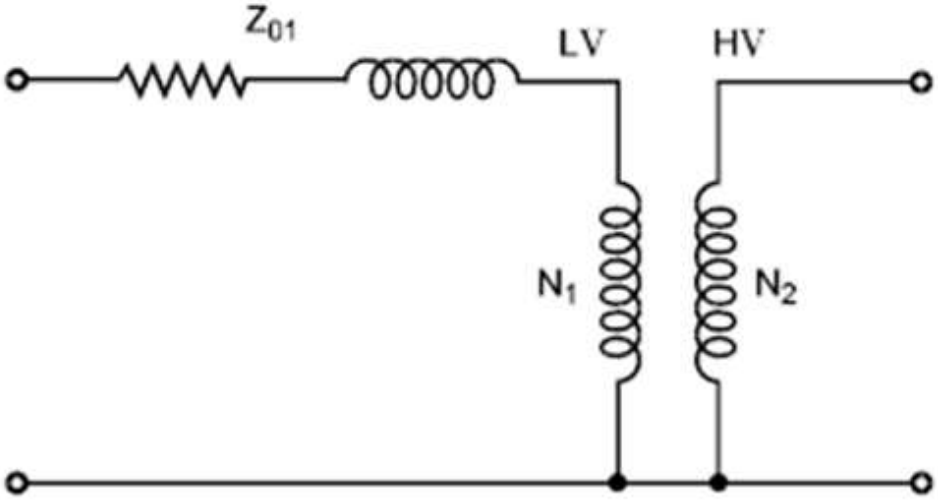
With common S_B and $V_{B,H} = tV_{B,L}$,

$$Z_{B,H} = t^2 Z_{B,L}, \qquad \frac{Z_H}{Z_{B,H}} = \frac{t^2 Z_L}{t^2 Z_{B,L}} = \frac{Z_L}{Z_{B,L}}.$$

Thus the transformer leakage impedance has the same p.u. value on both sides. It remains in the equivalent diagram; the *ideal ratio element* can be removed. For three-phase transformers, propagate line-voltage bases using the rated line-voltage ratio, which includes connection effects; do not automatically substitute a winding turns ratio for it. Vector-group phase shifts and zero-sequence connections still matter in their appropriate studies.

Transformer equivalent

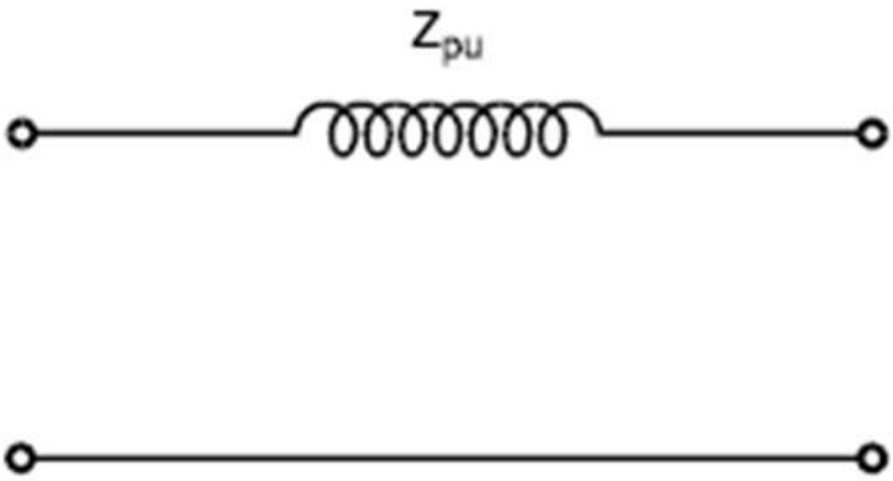
Class-slide anchor: Per-unit, p. 14.



The ideal turns ratio disappears only with consistent voltage bases on the two sides. Leakage impedance remains.

Referred impedance

Class-slide anchor: Per-unit, p. 15.



A single per-unit series impedance can replace the referred transformer equivalent.

A reliable base-selection workflow

Choose one system MVA base. Choose voltage base at one bus. Propagate voltage bases through transformer rated voltage ratios. Calculate each local Z_B, I_B. Convert every nameplate or ohmic impedance to that local common base. Then draw the impedance/sequence network and combine values. At the end, multiply p.u. current by the base current *at the fault bus*.

3 Every per-unit class example

3.1 20 kVA single-phase transformer: 2000/200 V

Class-slide anchor: Per-unit part pp. 10–15.

Leakage $X = 0.8\ \Omega$ referred to the 200 V side:

$$Z_{B,L} = \frac{200^2}{20000} = 2\ \Omega, \qquad \boxed{X_{\text{pu}} = 0.8/2 = 0.4\ \text{p.u.} = 40\%}.$$

On the HV side $t = 10$, so $X_H = 0.8(10)^2 = 80\ \Omega$ and $Z_{B,H} = 2000^2/20000 = 200\ \Omega$. Again $80/200 = 0.4$ p.u. The large p.u. value is the given arithmetic; do not replace it by a guessed typical reactance.

3.2 Class Example 2.6: line impedance on 100 MVA, 132 kV

Class-slide anchor: Per-unit part pp. 17–18.

For $Z = 30 + \text{j}110\ \Omega$/phase,

$$Z_B = 132^2/100 = 174.24\ \Omega, \qquad \boxed{Z_{\text{pu}} = 0.172176 + \text{j}0.631313}.$$

The source's $0.172 + \text{j}0.631$ is this result rounded. Base voltage is line-to-line and MVA is total three-phase; Z is per phase.

3.3 Class Example 2.7: generator reactance on a new base

Class-slide anchor: Per-unit part pp. 19–20.

Given generator rating 30 MVA, 11 kV and $X_{old} = 0.2$ p.u.; requested base 50 MVA, 33 kV:

$$\boxed{X_{new} = 0.2\frac{50}{30}\left(\frac{11}{33}\right)^2 = 0.037037\ \text{p.u.}}.$$

This is a change to an arbitrarily requested voltage base *on the same physical side*. It does not claim the generator physically operates at 33 kV. If an actual 11/33 kV transformer were being crossed with bases following its ratio, the p.u. generator reactance would remain unchanged by that crossing, apart from any MVA-base change. Confirm by ohms: $X = 0.2(11^2/30) = 0.806667\ \Omega$, new $Z_B = 33^2/50 = 21.78\ \Omega$.

3.4 Balanced system: 30 MVA, 12 kV, given impedance

Class-slide anchor: Per-unit part pp. 21–24.

Given apparent-power magnitude $S = 30$ MVA, $V_{LL} = 12$ kV and $Z = 1.5 + \mathrm{j}2\ \Omega$/phase; initial bases 50 MVA, 10 kV:

$$I_B = \frac{50}{\sqrt{3}(10)} = 2.886751\ \text{kA}, \quad Z_B = 10^2/50 = 2\ \Omega,$$

$$\boxed{|V_{\text{pu}}| = 1.2}, \quad |I| = \frac{30}{\sqrt{3}(12)} = 1.443376\ \text{kA}, \quad \boxed{|I_{\text{pu}}| = 0.5}, \quad \boxed{Z_{\text{pu}} = 0.75 + \mathrm{j}1.0}.$$

$|Z_{\text{pu}}| = 1.25$. On 100 MVA, 20 kV,

$$\boxed{Z_{\text{pu,new}} = (0.75 + \mathrm{j}1)\frac{100}{50}\left(\frac{10}{20}\right)^2 = 0.375 + \mathrm{j}0.5}.$$

New $Z_B = 4\ \Omega$, providing the direct check $(1.5 + \mathrm{j}2)/4$.

Do not infer more than these data specify

No power factor is given, so the current result is a *magnitude*, not a fully known phasor. The supplied impedance cannot also be the sole load impedance consuming the stated 30 MVA at 12 kV: that would require $|Z_{load}| = V_{LL}^2/|S| = 4.8\ \Omega$, rather than 2.5 Ω. Treat Z as separately supplied component/network impedance. This avoids a false attempt to force $V/I = Z$ for unrelated quantities.

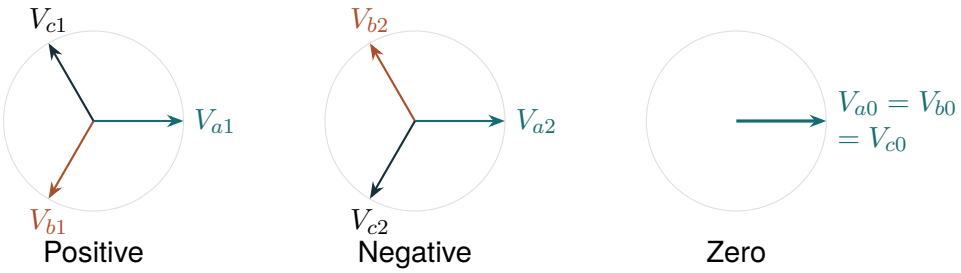
4 Symmetrical components: what the transform means

Class-slide anchor: Symmetrical-components part pp. 1–15, 26.

4.1 Decompose an unbalanced set into three simple patterns

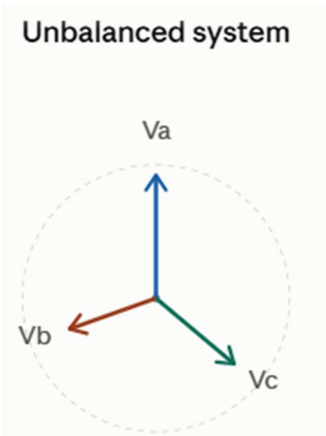
Any three phase phasors at the *same frequency* can be uniquely written as a sum of positive-, negative- and zero-sequence sets. This is an algebraic decomposition, not three physically separate power systems. Independent sequence circuits are obtained only when the underlying linear network is sequence decoupled, as in the usual balanced/transposed model.

Positive sequence has equal magnitudes, 120° separation and phase order a-b-c. Negative sequence has equal magnitudes, 120° separation and order a-c-b. Zero sequence has three identical, in-phase phasors; it is symmetric but is not a 120° balanced rotating set.



An unbalanced phasor set

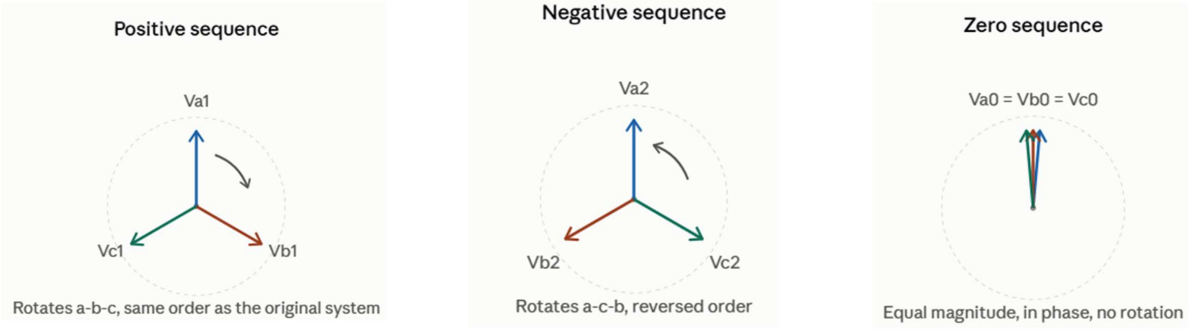
Class-slide anchor: Sequence, p. 5.



Unequal magnitudes or unequal angular separations require component decomposition.

Three sequence phasor sets

Class-slide anchor: Sequence, p. 6.



Read the phase labels: positive a-b-c, negative a-c-b, zero all coincident. The main text fixes a=1 angle 120 degrees.

Define

$$a = -\tfrac{1}{2} + \mathrm{j}\tfrac{\sqrt{3}}{2}, \quad a^2 = -\tfrac{1}{2} - \mathrm{j}\tfrac{\sqrt{3}}{2}, \quad a^3 = 1, \quad 1 + a + a^2 = 0, \quad a^{-1} = a^2.$$

Multiplication by a rotates a phasor 120° anticlockwise. Positive sequence obeys $V_{b1} = a^2 V_1$, $V_{c1} = aV_1$; negative sequence obeys $V_{b2} = aV_2$, $V_{c2} = a^2 V_2$. These relations fix the signs and positions in the matrices.

4.2 From sequence quantities back to actual phase quantities

Each phase is the sum of its three component contributions:

$$V_a = V_0 + V_1 + V_2, \quad V_b = V_0 + a^2 V_1 + aV_2, \quad V_c = V_0 + aV_1 + a^2 V_2.$$

Thus

$$\boxed{\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix}}_{T} \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix}}.$$

The same transform applies to currents and other three-phase phasor sets. Always state the sequence-vector order; textbooks also use $(1, 2, 0)$, which rearranges the matrix.

4.3 Derive the inverse transformation

Add the three phase equations. Since $1 + a + a^2 = 0$, the positive and negative contributions cancel, leaving $V_a + V_b + V_c = 3V_0$. Similarly, $V_a + aV_b + a^2 V_c = 3V_1$ and $V_a + a^2 V_b + aV_c = 3V_2$.

$$\boxed{\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}}.$$

Verify $T^{-1}T = I$ using $a^3 = 1$ and $1+a+a^2 = 0$. In a balanced positive-sequence set, only V_1 remains. A reverse balanced set has only V_2; equal in-phase phasors have only V_0. An arbitrary unbalanced set need not contain all three: a three-wire set with zero current sum has $I_0 = 0$ even when unbalanced.

4.4 Physical consequences that make the decomposition useful

$$I_a + I_b + I_c = 3I_0.$$

If phase currents are all referenced outward from a bus, the return-path current balances this sum with the appropriate opposite sign. Zero-sequence current therefore needs a return path: neutral,

earth, another conductor, or circulating winding paths as allowed by the equipment. It is not limited to physical earth. Zero-sequence *voltage* can exist without a conducting earth-current path, as in neutral displacement.

Line-to-line voltage cancels zero sequence: $V_{ab} = V_a - V_b$. Consequently zero-sequence voltage cannot be recovered from just line-to-line voltages; phase-to-reference information is needed. A balanced positive-sequence generator produces no negative/zero internal EMF in the elementary model, yet these *currents* can flow during faults because the fault conditions connect the networks and the positive source drives them.

An ideal symmetrical-component power identity, useful as a check, is

$$S_{3\phi} = V_a I_a^* + V_b I_b^* + V_c I_c^* = 3(V_0 I_0^* + V_1 I_1^* + V_2 I_2^*).$$

For a rotating machine, negative-sequence currents produce a reverse rotating stator field, seen at twice synchronous electrical frequency by its rotor, causing heating. This explains why unbalance can be damaging even without a very large positive-sequence current. Converter-connected sources may have controls and current limits that require a more detailed model than a fixed EMF behind reactance.

5 All five class symmetrical-component examples

5.1 Balanced normal phase order

Class-slide anchor: Symmetrical-components part pp. 16–17.

$V_a = 100\angle 0^\circ$, $V_b = 100\angle -120^\circ = 100a^2$, $V_c = 100\angle 120^\circ = 100a$ V:

$$V_0 = \tfrac{100}{3}(1 + a^2 + a) = 0, \quad V_1 = \tfrac{100}{3}(1 + a^3 + a^3) = 100, \quad V_2 = \tfrac{100}{3}(1 + a + a^2) = 0.$$

$$\boxed{V_0 = 0, \quad V_1 = 100\angle 0^\circ \text{ V}, \quad V_2 = 0}.$$

5.2 Balanced reverse phase order

Class-slide anchor: Symmetrical-components part pp. 18–19.

Now $V_a = 100$, $V_b = 100a$, $V_c = 100a^2$ V. Substitute into the inverse transform:

$$\boxed{V_0 = 0, \quad V_1 = 0, \quad V_2 = 100\angle 0^\circ \text{ V}}.$$

This is balanced negative sequence, not a mixed set merely because it has reverse order.

5.3 Three identical in-phase voltages

Class-slide anchor: Symmetrical-components part pp. 20–21.

$V_a = V_b = V_c = 60\angle 0^\circ$ V. Hence $V_0 = (60 + 60 + 60)/3 = 60$, while V_1 and V_2 vanish through $1 + a + a^2 = 0$:

$$\boxed{V_0 = 60\angle 0^\circ \text{ V}, \quad V_1 = V_2 = 0}.$$

Line-to-line voltages are all zero, despite nonzero phase-to-reference voltages.

5.4 Unequal phase magnitudes

Class-slide anchor: Symmetrical-components part pp. 22–23.

$V_a = 100$, $V_b = 80a^2$, $V_c = 60a$ V. Convert to rectangular form or simplify the rotations algebraically:

$$\begin{aligned} V_0 &= \tfrac{1}{3}(100 + 80a^2 + 60a) = 10 - \text{j}5.77350, \\ V_1 &= \tfrac{1}{3}(100 + 80 + 60) = 80, \\ V_2 &= \tfrac{1}{3}(100 + 80a + 60a^2) = 10 + \text{j}5.77350. \end{aligned}$$

$$\boxed{V_0 = 11.5470\angle -30^\circ \text{ V}, \quad V_1 = 80\angle 0^\circ \text{ V}, \quad V_2 = 11.5470\angle 30^\circ \text{ V}}.$$

Check phase a: $(10 - \text{j}5.7735) + 80 + (10 + \text{j}5.7735) = 100$. Check the other two by the forward transformation; cancellation in phase a alone is insufficient to validate the entire set.

5.5 Given sequence currents, recover phase currents

Class-slide anchor: Symmetrical-components part pp. 24–25.

$I_0 = 2$, $I_1 = 10$, $I_2 = 4$ A, all at zero degrees:

$$I_a = 2 + 10 + 4 = 16, \quad I_b = 2 + 10a^2 + 4a = -5 - \text{j}5.19615,$$

$$I_c = 2 + 10a + 4a^2 = -5 + \text{j}5.19615.$$

$$\boxed{I_a = 16\angle 0^\circ \text{ A}}, \quad \boxed{I_b = 7.21110\angle -133.898^\circ \text{ A}}, \quad \boxed{I_c = 7.21110\angle 133.898^\circ \text{ A}}.$$

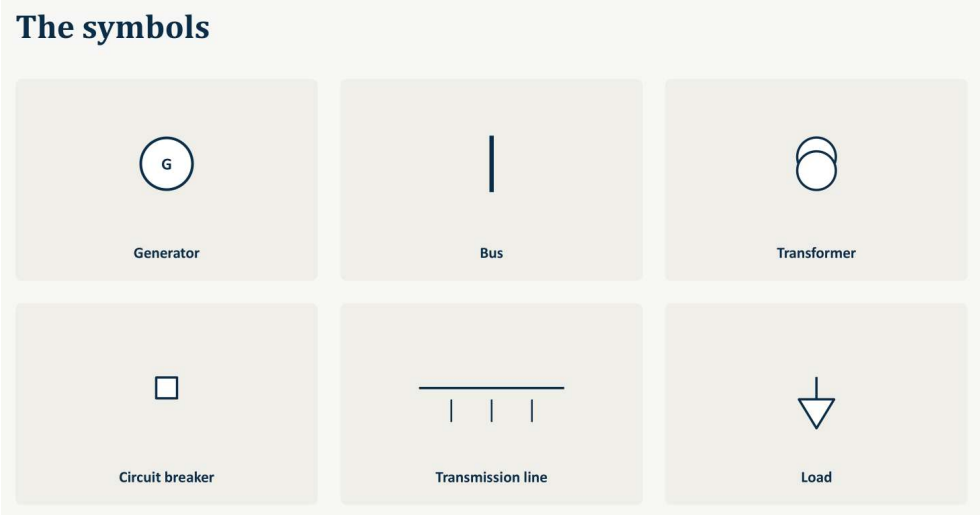
Current sum is $6 \text{ A} = 3I_0$, not zero. A physical implementation needs a permitted return path for that current.

6 Build sequence networks from real equipment

Class-slide anchor: Symmetrical-components part pp. 9–11, 14–15; fault-analysis part pp. 7–11.

One-line symbol key

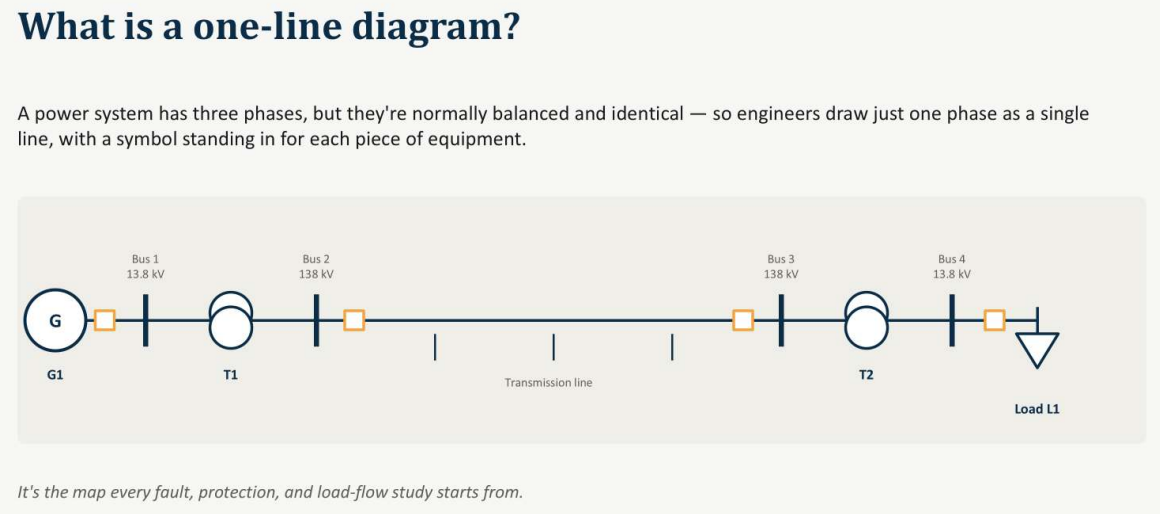
Class-slide anchor: Fault, p. 9.



Recognize generator, bus, transformer, circuit breaker, transmission line and load before drawing equivalent networks.

A class one-line system

Class-slide anchor: Fault, p. 8.



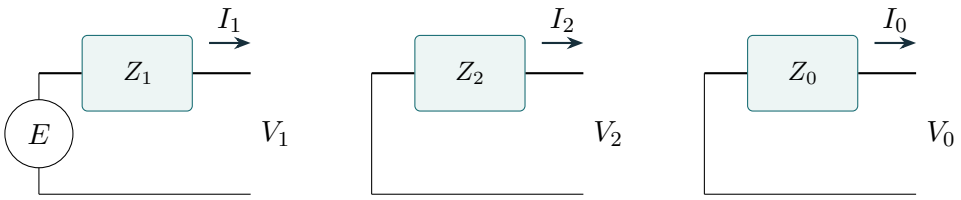
Generator, transformers, line, breakers, buses and load are represented by one phase of a balanced three-phase network.

6.1 The three fault-bus equivalents

For a balanced pre-fault system reduced to the fault bus,

$$\boxed{V_1 = E - Z_1 I_1}, \qquad \boxed{V_2 = -Z_2 I_2}, \qquad \boxed{V_0 = -Z_0 I_0}.$$

The minus signs follow the current direction into the fault. They are essential in LL/LLG calculations. Networks 2 and 0 have no independent internal source under the stated balanced-source model.



Functional Thevenin equivalents. Bottom nodes are sequence reference nodes. A drawn reference is not itself proof that a physical zero-sequence earth-return path exists.

6.2 Generator models and neutral impedance

Positive sequence uses internal EMF behind an appropriate reactance. Initially after a short circuit, use subtransient X_d'' if specified; later use transient X_d' or synchronous X_d for the corresponding time interval. The AC current envelope generally decays through these stages. Resistance and an inception-dependent decaying DC offset are omitted in elementary symmetrical RMS phasor calculations; the first asymmetrical peak is not just $\sqrt{2}$ times a steady-state fault current.

Negative sequence uses Z_2 with no balanced-source internal EMF. For a machine it need not equal Z_1; use the given data. Zero sequence uses $Z_{0,G}$ and the actual neutral connection. If neutral-to-earth impedance is Z_n, zero-sequence phase currents are equal and neutral current has magnitude/phasor sum $3I_0$. Its voltage drop is $3Z_nI_0$, so the per-phase zero network contains

$$\boxed{Z_{0,\text{branch}} = Z_{0,G} + 3Z_n}.$$

An ungrounded neutral blocks the simple conductive zero-sequence branch, but actual networks may still have capacitive earth paths. Do not count $3Z_n$ twice if the supplied Z_0 already includes neutral grounding.

6.3 Lines and transformers

A balanced/transposed line usually has $Z_2 = Z_1$ in the elementary model. Its Z_0 is different because same-phase zero-sequence currents return through earth/neutral and link different magnetic paths. Use measured/given zero-sequence data rather than guessing $Z_0 = Z_1$.

A transformer has approximately equal positive and negative leakage sequence impedances in the usual model; vector groups can introduce opposite phase shifts in the two sequences. Zero-sequence behaviour depends on winding connections, neutral grounding, magnetic core construction and external paths.

Connection	Elementary zero-sequence line-current behaviour
Grounded star / delta	Grounded-star side can support zero-sequence line current with corresponding delta circulation. It cannot appear as zero-sequence *line current* on the delta side.
Ungrounded star	No conductive neutral return in the elementary model: external zero-sequence line-current path is open.
Delta	Zero-sequence winding currents may circulate internally. Equal zero-sequence components cancel in delta line currents.
Grounded star / grounded star	Zero-sequence transfer may be possible when both returns and the transformer magnetic path permit it. Include neutral impedances and the specified zero-sequence transformer model.

For a delta, line current is the difference of adjacent winding currents. Equal circulating components give $I_{a0,\text{line}} = I_{ab0} - I_{ca0} = 0$, while winding current can remain nonzero. Saying "delta blocks zero sequence" is acceptable only if you mean zero-sequence current *in its external lines*.

6.4 Reduce the network at the fault bus

Deactivate independent voltage sources to compute sequence driving-point impedances, preserving their internal impedance. Combine series/parallel paths on the common base. Preserve connection-dependent zero-sequence topology. For a multi-bus study, network inversion or a bus-impedance matrix can provide the driving-point element; simple class examples are already reduced.

Additional example: grounding changes only the required branch

A star generator has $Z_1 = \text{j}0.2$, $Z_2 = \text{j}0.2$, $Z_{0,G} = \text{j}0.1$ p.u. and neutral reactance $Z_n = \text{j}0.05$ p.u. Its positive/negative values remain $\text{j}0.2$; its effective zero-sequence branch becomes $\text{j}(0.1 + 3 \times 0.05) = \text{j}0.25$. A solid LG fault gives $I_a = 3/(\text{j}0.65) = -\text{j}4.61538$ p.u., compared with $-\text{j}6$ for solid neutral grounding. A three-phase fault remains $I_a = 1/(\text{j}0.2) = -\text{j}5$ p.u. in this model.

7 Derive each shunt-fault connection

Class-slide anchor: Fault-analysis part pp. 1–18.

7.1 Fault types, causes and consequences

A shunt fault makes an unintended conducting path between phases and/or to earth. Causes include insulation deterioration, lightning/overvoltage flashover, damaged equipment, contamination, vegetation/animals and mechanical/human error. Arcing and high-resistance faults need not be literal metal-to-metal contact or an extreme current surge. They still can create dangerous conditions.

Balanced three-phase faults (LLL, or balanced LLLG) are symmetrical. LG, LL and LLG are unsymmetrical. Open-conductor/series faults are another category, recognised here but not calculated because the syllabus specifies shunt calculations only. Short circuits can heat and mechanically stress equipment, depress voltage, endanger people and disrupt stability. Analysis supports breaker-duty selection and protection studies.

Do not rank every fault by a universal severity rule

A three-phase fault is a common reference for breaker duty, but it is not always the highest current. A solid LG fault can exceed it if zero-sequence impedance is sufficiently low. The actual network, grounding and fault impedance determine the result.

7.2 Balanced three-phase fault: positive sequence only

For equal phase fault impedances Z_f, the fault is balanced; $I_0 = I_2 = 0$. Phase-a boundary condition is $V_a = Z_f I_a$. Since $V_a = V_1$ and $I_a = I_1$,

$$E - Z_1 I_1 = Z_f I_1 \quad \Rightarrow \quad \boxed{I_1 = \frac{E}{Z_1 + Z_f}}, \qquad I_a = I_1,\ I_b = a^2 I_1,\ I_c = a I_1.$$

For a solid fault $Z_f = 0$ and the fault phase voltages are zero in the ideal model. A balanced LLL fault without earth has the same positive-sequence result; no zero sequence is generated just by adding a balanced ground connection.

7.3 Single-line-to-ground fault on phase a

At the fault point, unfaulted-branch currents are $I_b = I_c = 0$, while $V_a = Z_f I_a$. Apply the inverse current transform:

$$I_0 = I_1 = I_2 = I_a/3.$$

Add the three sequence voltages:

$$V_a = V_0 + V_1 + V_2 = E - (Z_1 + Z_2 + Z_0) I_1 = 3 Z_f I_1.$$

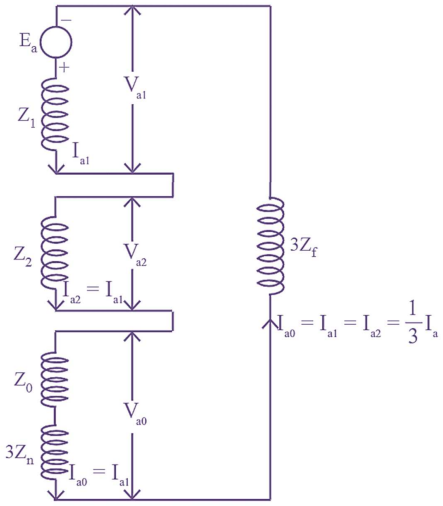
Thus

$$\boxed{I_0 = I_1 = I_2 = \frac{E}{Z_1 + Z_2 + Z_0 + 3Z_f}}, \qquad \boxed{I_a = \frac{3E}{Z_1 + Z_2 + Z_0 + 3Z_f}}.$$

The three sequence impedances and $3Z_f$ are in series. Z_f is the phase-to-earth fault impedance, not a neutral impedance; both may appear in the same problem, in different places. Fault-branch $I_b = I_c = 0$ does not mean all upstream load currents are zero in a loaded network.

LG series connection

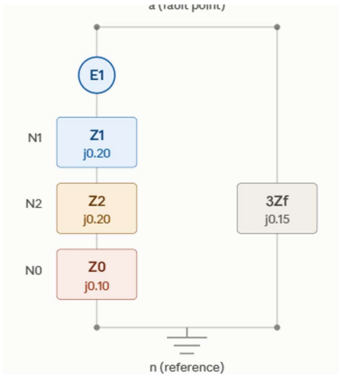
Class-slide anchor: Fault, p. 14.



All three sequence currents are equal. Neutral impedance 3Zn belongs inside Z0; the external phase fault appears as 3Zf, counted separately.

Finite-impedance LG network

Class-slide anchor: Fault, p. 25.



The return branch is 3Zf because the physical phase current is three times the sequence current.

7.4 Line-to-line fault between b and c

With impedance Z_f between the two phases,

$$I_a = 0, \quad I_b = -I_c, \quad V_b - V_c = Z_f I_b.$$

There is no return current, so $I_0 = 0$; then $I_2 = -I_1$. From the transform,

$$V_b - V_c = (a^2 - a)(V_1 - V_2), \quad I_b = (a^2 - a)I_1 = -\mathrm{j}\sqrt{3}I_1.$$

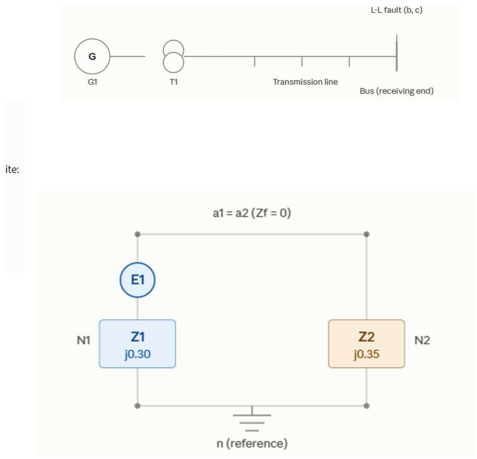
Therefore $V_1 - V_2 = Z_f I_1$. Substitute $V_1 = E - Z_1 I_1$ and $V_2 = -Z_2 I_2 = Z_2 I_1$:

$$\boxed{I_1 = \frac{E}{Z_1 + Z_2 + Z_f}}, \quad I_2 = -I_1,\ I_0 = 0; \qquad \boxed{I_b = -\mathrm{j}\sqrt{3} I_1,\ I_c = -I_b}.$$

For a solid fault $V_1 = V_2$. The positive and negative networks have common terminal voltage and opposing fault-directed currents: textbooks draw their terminals in parallel. The calculation's denominator is still $Z_1 + Z_2$, not $Z_1 \parallel Z_2$.

LL connection and one-line location

Class-slide anchor: Fault, p. 27.



The positive and negative ports share their fault voltage; their network currents have opposite signs. Zero sequence is absent.

7.5 Double-line-to-ground fault on b and c

Assume phases b, c are joined at a common node, connected to earth through Z_f. This is a *common ground impedance*; two separate phase-to-earth impedances require different boundary equations.

$$I_a = 0 \Rightarrow I_0 + I_1 + I_2 = 0, \qquad V_b = V_c.$$

Subtract the phase voltages to get $V_1 = V_2 = V$. Then $V_b = V_c = V_0 - V$, while $I_b + I_c = 3I_0$. Consequently

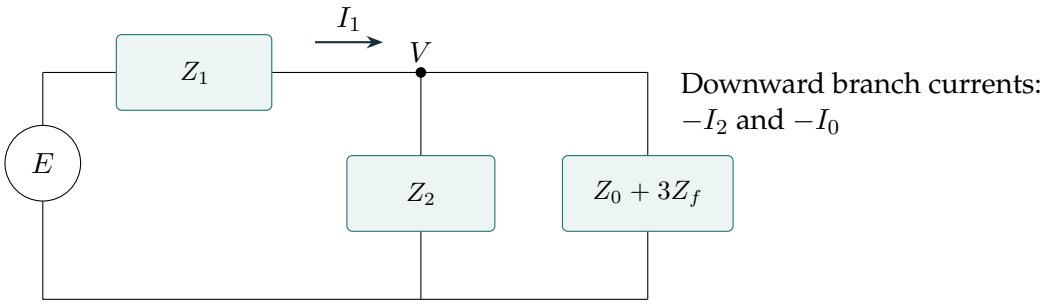
$$V_0 - V = 3Z_f I_0, \quad V_0 = -Z_0 I_0 \quad \Rightarrow \quad V = -(Z_0 + 3Z_f) I_0.$$

Also $V = -Z_2 I_2$. Since $I_1 = -(I_0 + I_2)$,

$$\boxed{I_1 = \frac{E}{Z_1 + \dfrac{Z_2(Z_0 + 3Z_f)}{Z_2 + Z_0 + 3Z_f}}},$$

$$\boxed{I_2 = -I_1 \frac{Z_0 + 3Z_f}{Z_2 + Z_0 + 3Z_f}}, \qquad \boxed{I_0 = -I_1 \frac{Z_2}{Z_2 + Z_0 + 3Z_f}}.$$

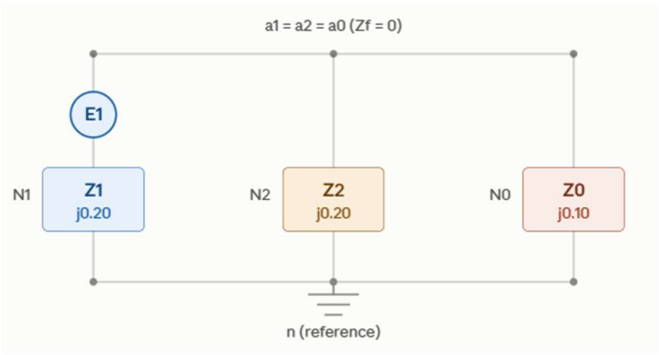
Transform to I_b, I_c; total earth-fault current is $I_b + I_c = 3I_0$. For a solid fault $Z_f = 0$, all three sequence terminal voltages are equal: $V_0 = V_1 = V_2$. The positive network feeds the parallel combination of negative and zero impedances. Negative/zero *physical sequence currents into the fault* have the opposite sign to the branch currents drawn out of the common sequence node.



LLG equivalent connection, with current-sign convention made explicit. For solid LLG use $Z_f = 0$. For LG the three impedances are in series; for LL the zero branch is absent.

Solid LLG connection

Class-slide anchor: Fault, p. 31.



Positive sequence feeds the parallel negative and zero networks. Use the corrected current signs in Section **??**; the diagram does not justify copying the erroneous source arithmetic.

7.6 A calculation workflow and powerful checks

Draw a one-line diagram; state fault type/location and fault impedance definition; select common bases; construct each equipment's sequence connections; reduce to E, Z_1, Z_2, Z_0 at the fault; apply boundary conditions; solve sequence currents; transform to phase currents/voltages; convert to actual values at that bus. Finally check

- LG: $I_b = I_c = 0$, $I_a = 3I_0$, $V_a = Z_f I_a$.
- LL: $I_a = 0$, $I_b = -I_c$, $I_0 = 0$, $V_b - V_c = Z_f I_b$.
- Solid LLG: $I_a = 0$, $V_b = V_c = 0$, $I_b + I_c = 3I_0$.
- Three phase: equal current magnitudes 120° apart, $I_0 = I_2 = 0$.

For symmetric short-circuit apparent power, $S_{sc} = \sqrt{3} V_{LL,\text{nominal}} |I_f|$. With $E = 1$ p.u., a reactance-only three-phase Thevenin model gives $S_{sc} = S_B / X_{1,\text{pu}}$. Nominal voltage is used for this duty convention; the bolted fault-point voltage itself is zero.

8 Every class fault calculation, with checks

8.1 Class case 1: solid three-phase fault, positive impedance 0.20 p.u.

Class-slide anchor: Fault-analysis part pp. 19–20.

With $E = 1$, $Z_1 = \text{j}0.20$,

$$I_1 = 1/(\text{j}0.20) = -\text{j}5 = 5\angle -90^\circ, \qquad I_0 = I_2 = 0.$$

$$\boxed{I_a = 5\angle -90^\circ}, \quad \boxed{I_b = 5\angle 150^\circ}, \quad \boxed{I_c = 5\angle 30^\circ} \quad \text{p.u.}$$

Multiplication by a^2 rotates I_a another -120°: -210° is the same angle as 150°. A bare magnitude of 5 does not describe the phase relationships.

8.2 Class case 2: three-phase current at an 11 kV bus

Class-slide anchor: Fault-analysis part pp. 21–22.

$S_B = 100$ MVA, $V_B = 11$ kV, $Z_1 = \text{j}0.25$ p.u. and $E = 1$:

$$|I_f| = |1/(\text{j}0.25)| = 4 \text{ p.u.}, \quad I_B = \frac{100}{\sqrt{3}(11)} = 5.248639 \text{ kA},$$

$$\boxed{|I_f| = 4(5.248639) = 20.9946 \text{ kA}}.$$

The diagram on the solution slide includes extraneous sequence branches and a different positive label; the question's $Z_1 = \text{j}0.25$ is the value that gives its stated 4 p.u. result. Only the positive network is needed for this balanced fault. Associated short-circuit level is $100/0.25 = 400$ MVA.

8.3 Class case 3a: solid LG fault

Class-slide anchor: Fault-analysis part pp. 23–24.

Given $Z_1 = \text{j}0.20$, $Z_2 = \text{j}0.20$, $Z_0 = \text{j}0.10$, $E = 1$, solid fault on a:

$$\boxed{I_0 = I_1 = I_2 = \frac{1}{\text{j}0.50} = -\text{j}2 \text{ p.u.}}, \quad \boxed{I_a = -\text{j}6 \text{ p.u.}, \quad I_b = I_c = 0}.$$

The sequence voltages are

$$V_1 = 1 - (\text{j}0.2)(-\text{j}2) = 0.6, \quad V_2 = -(\text{j}0.2)(-\text{j}2) = -0.4, \quad V_0 = -(\text{j}0.1)(-\text{j}2) = -0.2.$$

Check $V_a = 0.6 - 0.4 - 0.2 = 0$. For the same bus, three-phase current is 5 p.u.; this LG current is 6 p.u. Thus a three-phase fault is not always the largest current.

8.4 Class case 3b: the same LG fault with impedance

Class-slide anchor: Fault-analysis part pp. 23, 25.

Now $Z_f = \text{j}0.05$ p.u.:

$$I_0 = I_1 = I_2 = \frac{1}{\text{j}(0.2 + 0.2 + 0.1 + 3 \times 0.05)} = -\text{j}1.538462 \text{ p.u.},$$

$$\boxed{I_a = -\text{j}4.615385 \text{ p.u.}, \quad I_b = I_c = 0}.$$

$V_1 = 0.6923077$, $V_2 = -0.3076923$, $V_0 = -0.1538462$, so

$$V_a = 0.2307692 = Z_f I_a = (\text{j}0.05)(-\text{j}4.615385).$$

Fault impedance reduces current here. Its factor of three results from the fault phase current being three times each sequence current.

8.5 Class case 4: LL fault at the receiving end

Class-slide anchor: Fault-analysis part pp. 26–29.

All data already share a common p.u. base. Generator $X_{1G} = 0.15$, $X_{2G} = 0.20$; transformer $X_{1T} = X_{2T} = 0.10$; line $X_{1L} = X_{2L} = 0.05$. The fault is solid between b, c and $E = 1$.

$$Z_1 = \text{j}(0.15 + 0.10 + 0.05) = \text{j}0.30, \quad Z_2 = \text{j}(0.20 + 0.10 + 0.05) = \text{j}0.35.$$

$$\boxed{I_1 = -\text{j}1.538462, \quad I_2 = +\text{j}1.538462, \quad I_0 = 0} \quad \text{p.u.}$$

$$I_b = -\text{j}\sqrt{3} I_1 = -2.664694, \qquad I_c = +2.664694,$$

$$\boxed{I_a = 0, \quad I_b = 2.664694\angle 180^\circ, \quad I_c = 2.664694\angle 0^\circ} \quad \text{p.u.}$$

Check $I_b + I_c = 0$. $V_1 = 1 - 0.3(1.538462) = 0.538462$; $V_2 = -(\text{j}0.35)(+\text{j}1.538462) = 0.538462$. Hence $V_b = V_c = -0.538462$ p.u. under this reference: the faulted phases have equal voltages, not necessarily zero voltage to earth.

8.6 Class case 5: LLG, with the slide sign error corrected

Class-slide anchor: Fault-analysis part pp. 30–33.

$Z_1 = \text{j}0.20$, $Z_2 = \text{j}0.20$, $Z_0 = \text{j}0.10$, $E = 1$; solid fault on b, c to earth. The parallel part is

$$Z_2 \parallel Z_0 = \frac{(\text{j}0.20)(\text{j}0.10)}{\text{j}0.30} = \text{j}0.0666667.$$

$$I_1 = \frac{1}{\text{j}0.2666667} = -\text{j}3.75, \qquad V = V_1 = 1 - (\text{j}0.20)(-\text{j}3.75) = 0.25.$$

From $V_2 = -Z_2 I_2$ and $V_0 = -Z_0 I_0$,

$$\boxed{I_2 = -\frac{0.25}{\text{j}0.20} = +\text{j}1.25}, \qquad \boxed{I_0 = -\frac{0.25}{\text{j}0.10} = +\text{j}2.50} \quad \text{p.u.}$$

Now transform:

$$\begin{aligned} I_a &= \text{j}2.5 - \text{j}3.75 + \text{j}1.25 = \boxed{0}, \\ I_b &= \text{j}2.5 + a^2(-\text{j}3.75) + a(\text{j}1.25) = \boxed{-4.330127 + \text{j}3.75}, \\ I_c &= \text{j}2.5 + a(-\text{j}3.75) + a^2(\text{j}1.25) = \boxed{4.330127 + \text{j}3.75}. \end{aligned}$$

Thus $|I_b| = |I_c| = 5.72822$ p.u. and the earth current is $I_b + I_c = \boxed{\text{j}7.5 = 3I_0}$ p.u. Also $V_0 = V_1 = V_2 = 0.25$, giving $V_b = V_c = 0$ and $V_a = 0.75$ p.u.

The physical check exposes the arithmetic error

Slide p. 32 writes $I_2 = -0.25/(\text{j}0.20)$ but evaluates it as $-\text{j}1.25$. Since $1/\text{j} = -\text{j}$, the leading minus makes the result *positive* imaginary. Its I_0 has the same error. Slide p. 33 then obtains $I_a = -\text{j}7.5$, contradicting the stated unfaulted phase-a condition $I_a = 0$. The correct 7.5 p.u. value is the magnitude of the total earth current, not current in phase a.

8.7 Class case 6: limiting three-phase current with a reactor

Class-slide anchor: Fault-analysis part pp. 34–35.

A 20 MVA, 10 kV alternator has $X_d'' = 5\% = 0.05$ p.u. Find additional series reactance to limit a terminal three-phase short circuit to eight times full-load current. Use the machine's ratings as bases and $E = 1$.

$$I_B = \frac{20}{\sqrt{3}(10)} = 1.154701\ \text{kA}, \quad I_{\max} = 8I_B = 9.237604\ \text{kA}, \quad X_{\text{total,pu}} = 1/8 = 0.125.$$

$$\boxed{X_{\text{ext,pu}} = 0.125 - 0.05 = 0.075 = 7.5\%}.$$

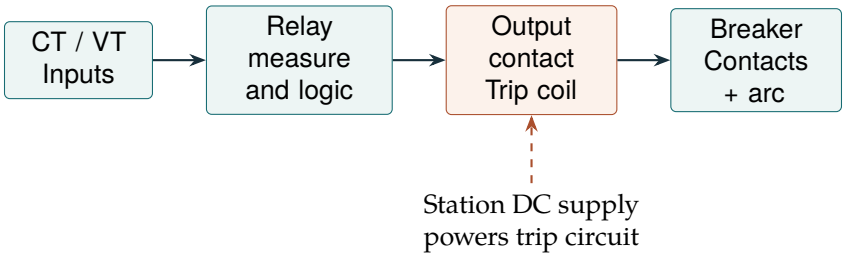
$Z_B = 10^2/20 = 5\ \Omega$, hence $\boxed{X_{\text{ext}} = 0.075(5) = 0.375\ \Omega/phase}$. At 50 Hz this corresponds to $L = X/\omega = 1.19366$ mH/phase. The current multiple and reactance percentage must use the same machine base; confusing a 5% reactance with 5 p.u. reverses the scale by a factor of 100.

9 Protection: sensing, selectivity and backup

Class-slide anchor: Protection part pp. 1–14, 31, 37.

9.1 Why protection is needed and what each device does

Protection reduces the duration and extent of abnormal operation. It cannot prevent every fault. A **CT** supplies scaled current, a **VT/PT** supplies scaled voltage, and the **relay** measures and evaluates them. The relay's trip output energises the breaker trip circuit from a control supply. A latch releases stored spring/pneumatic/hydraulic energy; the **circuit breaker** separates power contacts and interrupts the current. An alarm can be issued without a trip when the logic requires it.



Total clearing time includes measurement/relay decision, any intentional delay, output/auxiliary operation, breaker opening and arcing to final current interruption. "Instantaneous relay" means no intentional time delay; it still has finite operating time. The relay does not directly switch thousands of amperes through its small output contacts.

Relay and breaker recognition

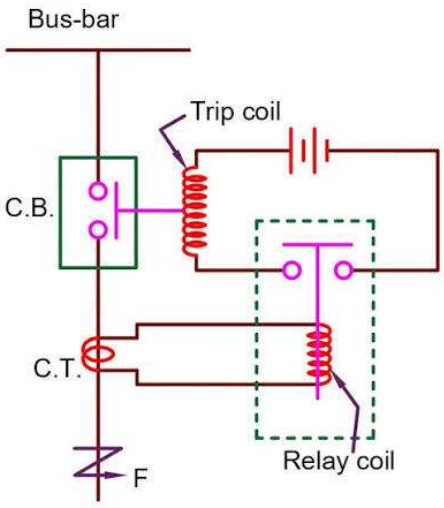
Class-slide anchor: Protection, p. 5.



The relay decides; the breaker interrupts the power circuit. They have different current ratings and functions.

CT-relay-trip-coil circuit

Class-slide anchor: Protection, p. 16.



Follow the low-power sensing circuit separately from the DC trip supply. A relay contact energizes the breaker trip coil.

9.2 Requirements and their tradeoffs

Requirement	Meaning and useful test
Selectivity	Isolate the faulted section while keeping healthy supply where possible. A fault near one feeder end should not unnecessarily trip the whole substation.
Speed	Clear quickly enough to limit damage and preserve stability. Some deliberate delay is needed for graded backup.
Sensitivity	Detect the minimum credible fault, including weak-source/far-end conditions. Pickup still must avoid permissible load and transients.
Stability/security	Avoid unwanted operation for external faults, allowable load, CT error, inrush or other non-trip conditions relevant to the scheme.
Dependability	Operate for faults the scheme is assigned to clear. Reliability includes both dependable operation and security against false trips.
Simplicity/economy	Use a maintainable, testable arrangement suited to the protected equipment and required reliability.

The slides emphasise four S's: selectivity, stability, speed and sensitivity. They must be interpreted within a declared protection zone; a backup relay *is intended* to clear a fault outside its primary zone if primary protection fails.

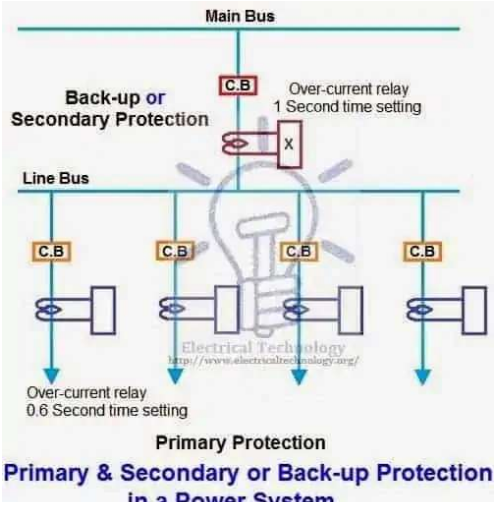
9.3 Primary protection, local backup and remote backup

Primary/main protection is assigned to clear faults in its protected zone with the intended speed/selectivity. It can include intentional delay; "primary" is a responsibility, not a guarantee of instantaneous action. **Backup protection** acts if a main relay, trip circuit, DC supply, communications path or breaker fails. It normally waits long enough to allow the primary scheme to work.

Local backup may be an independent relay system or breaker-failure scheme at the same station. Remote backup is provided by relays/breakers at another station; it can cover local failure but may remove more healthy equipment and may be less sensitive because of network infeed. Independent main schemes can be used to reduce common failure modes.

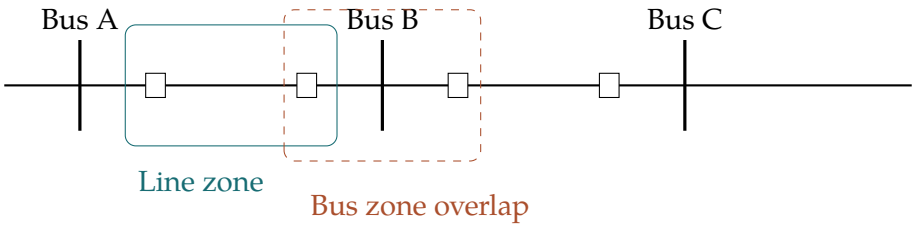
Primary and backup arrangement

Class-slide anchor: Protection, p. 6.



The feeder device isolates its local fault first. The upstream device supplies delayed backup; settings depend on the actual installation.

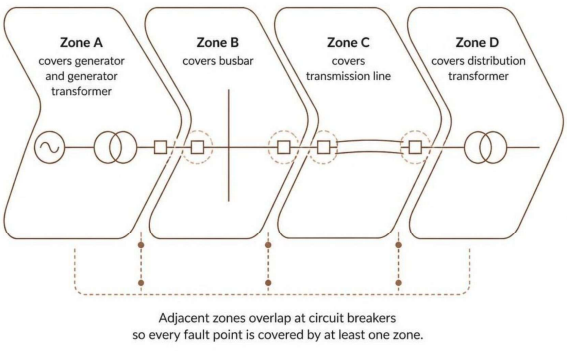
Zones are associated with equipment such as generators, transformers, buses and lines. CT locations define many unit-protection boundaries. Adjacent zones overlap around breakers so a fault at a boundary can be cleared. This equipment-zone overlap is different from the numbered reach zones of one distance relay.



Conceptual overlapping protection boundaries. Practical CT placements, breaker-failure logic and zone assignments must be specified for an actual station.

Overlapping protection zones

Class-slide anchor: Protection, p. 31.



CT locations define zone boundaries around breakers. This equipment-zone sketch is distinct from distance-relay reach zones.

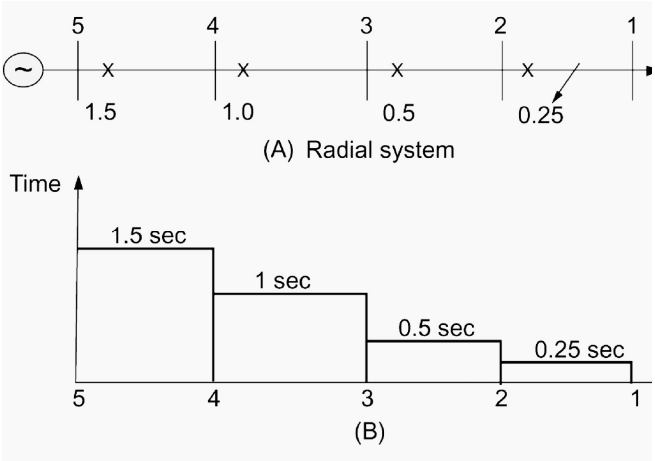
9.4 Relay coordination by time and current

In a radial feeder, all upstream relays can see a downstream fault. **Time grading** sets the nearest downstream relay to trip first; successive upstream relays provide delayed backup. Definite-time grading is simple but can make source-end fault clearing slow. Inverse-time curves permit shorter times for higher current, yet coordination must be checked across the credible fault-current range.

Current grading uses decreasing available fault current with distance in a simple radial network. Choose pickup above maximum permissible load and below the minimum fault current that the relay must detect, with required margins. Similar fault magnitudes at adjacent positions, multiple sources and changing topology limit pure current discrimination. Time-current grading combines pickup and delay settings.

Time grading

Class-slide anchor: Protection, p. 10.



The source illustration gives example delays. Coordination needs a margin covering breaker time, relay error and overtravel, not these exact universal settings.

Current grading

Class-slide anchor: Protection, p. 11.

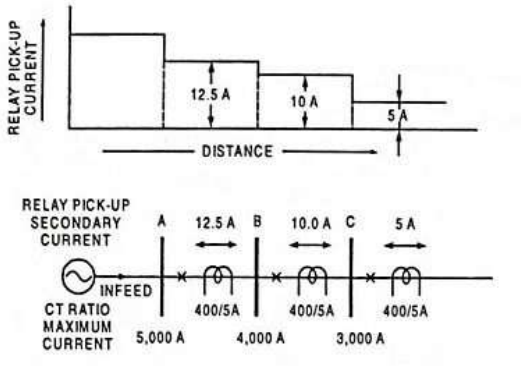


Fig. 14.9. *Current Graded Protection Applied To a Radial Feeder*

Compare pickup and fault current along a radial feeder. Current grading alone fails where adjacent fault currents are too similar.

Additional example: a concrete coordination margin

For one downstream fault, downstream relay operating time is 0.20 s and its breaker interruption time after trip output is 0.06 s. If the specified allowance for overshoot/tolerances is 0.10 s, upstream relay operating time must be at least $0.20 + 0.06 + 0.10 = 0.36$ s under this simplified coordination rule. This is a 0.16 s difference between relay times, not a universal setting. Repeat at relevant fault locations/current levels.

9.5 Breaker failure and interpreting slide timing values

Breaker-failure logic starts after a valid trip initiation and checks whether current and/or breaker status indicates failure to interrupt. A timed failure output can retrip or trip adjacent/upstream breakers to remove every source feeding the fault. Reset criteria avoid wider tripping after successful clearance. A breaker auxiliary contact alone and current alone each have limitations; actual schemes specify their supervision.

The slides give illustrative values: 3–10 cycles for a fault-clearing chain, 15–17 cycles for a breaker-failure timer, 0.2–0.5 s coordination intervals, and a 0.4 s electromechanical coordination example. They are not universal rules. At **50 Hz**, one cycle is 20 ms: 0.4 s is **20 cycles**, not the 24 cycles printed in the slides (24 cycles corresponds to 60 Hz). CT error/saturation is not automatically a separate fixed 12-cycle delay that is added to every fault. Specify the actual relay, breaker, tolerances and coordination assumptions.

9.6 Equipment applications and context from the slides

Transformer and generator differential protection are typical unit schemes. Transformer overcurrent can provide backup; a Buchholz gas/oil-motion relay applies to appropriate oil-filled conservator arrangements and detects particular internal abnormalities, so it is not a generic backup substitute for every transformer fault. Generator protection may include negative sequence, reverse power, loss of excitation and stator earth-fault functions. Lines may use distance, directional OC, line differential or communication-assisted schemes depending on their application.

Numerical IEDs can combine protection, metering, recording, control and communication. The slides' claims of 99.99% availability or immunity to network changes are aspirations/overstatements, not guarantees from selecting four relay principles. Adaptive settings and data-based fault detection

are contextual developments; dependable, tested protection and secure configuration remain essential.

10 The four protection principles

Class-slide anchor: Protection part pp. 13–38.

10.1 Overcurrent: magnitude plus a timing rule

An OC element picks up when measured current exceeds I_{pickup}. It may operate with no intentional delay (instantaneous, device 50), after a preset delay (definite time), or after a current-dependent delay (inverse-time, commonly device 51). A high-set instantaneous element supplements a delayed element when its reach can be made selective.

Define current multiple / plug-setting multiplier (PSM):

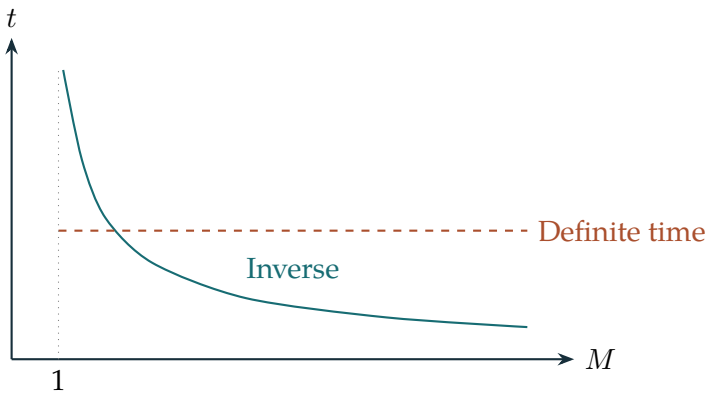
$$M = \frac{I_{\text{measured,secondary}}}{I_{\text{pickup,secondary}}} = \frac{I_{\text{primary}}}{I_{\text{primary,pickup}}}.$$

CT ratio and pickup must be applied consistently. A 400/5 A CT has ratio 80. Pickup 5 A secondary means 400 A primary; a 2000 A primary fault gives 25 A secondary and $M = 5$. "Pickup 100%" depends on the relay's nominal-current base; it is not 100% of fault current.

One widely used standard-inverse educational curve is

$$\boxed{t = \text{TMS}\frac{0.14}{M^{0.02} - 1}}, \qquad M > 1.$$

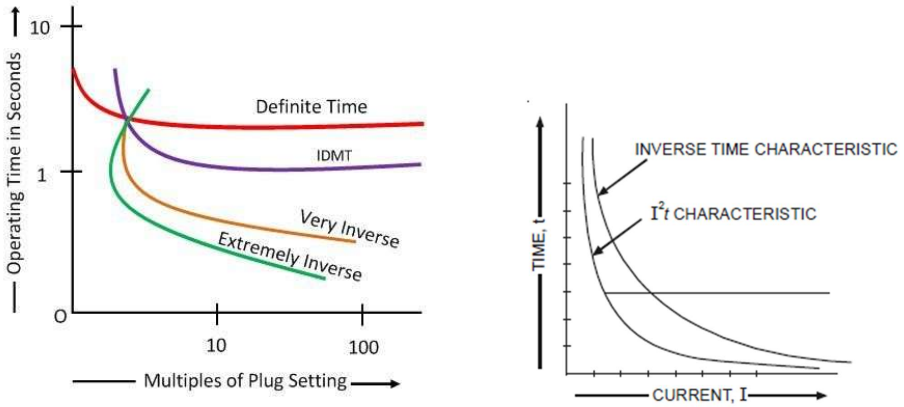
Other inverse shapes use different coefficients/exponents (for example very inverse $13.5/(M-1)$ and extremely inverse $80/(M^2-1)$, with the appropriate time multiplier). Manufacturer implementations and reset logic must be specified; one time-dial number is not interchangeable between all relay families. The curve defines relay operating time, not total breaker clearing time. The coefficients are cross-checked against the linked SEL technical manual in the references.



Conceptual time-current characteristics, not a calibrated manufacturer curve. Below pickup the element does not trip under this basic rule.

Overcurrent characteristic family

Class-slide anchor: Protection, p. 17.



Characteristic of Various Overcurrent Relay

Circuit Globe

Definite time, inverse, very inverse and extremely inverse curves describe operating time versus current multiple. Instantaneous still has finite operating time.

Additional example: current multiple and inverse time

For the 400/5 A CT above, choose 5 A secondary pickup and TMS $= 0.1$. At 2000 A primary, $M = 5$ and $t = 0.1(0.14)/(5^{0.02} - 1) = 0.427972$ s. At 4000 A, $M = 10$ gives $t = 0.297060$ s. Higher current gives shorter relay time. Add the relevant output/breaker delays to estimate total clearing time.

OC is well suited to many radial feeders. It does not directly identify direction or compare boundary currents. Weak-source far-end faults and significant distributed generation can complicate magnitude-based selectivity.

10.2 Directional: determine which way a fault lies

A directional element compares current phase with a *polarising/reference* voltage or another suitable reference. It distinguishes forward and reverse relative to the installed CT/VT polarity and chosen relay characteristic. There is no universal rule that every fault reverses normal power flow. A relay may deliberately trip for *forward* faults while blocking reverse faults.

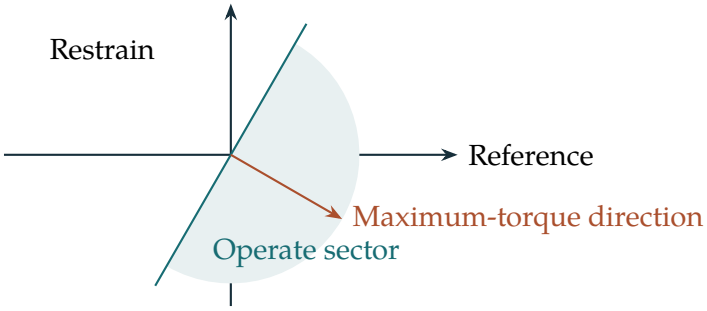
For an elementary voltage-current induction model, operating torque may be written

$$T_{op} = K|V||I|\cos(\theta - \tau) - T_{restraint},$$

where $\theta = \angle V - \angle I$ under the declared convention and τ is the maximum-torque/characteristic angle. The actual sign convention is relay specific. Directional OC combines directional permission with current pickup/timing:

$$\text{Trip} = (\text{forward permission}) \text{ AND } (\text{OC condition/timer complete}).$$

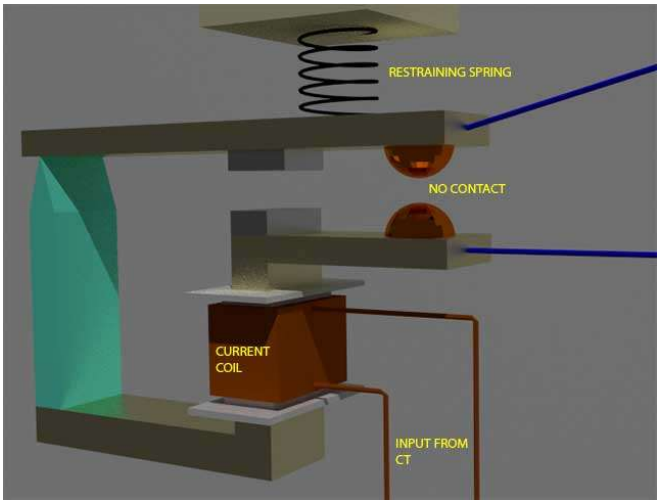
A directional element is therefore not simply an ammeter and not necessarily a complete fault detector on its own. Polarising voltage can be weak for a close-in fault; suitable cross/memory/sequence polarisation helps practical implementations.



Illustrative half-plane directional characteristic. Axes and angle reference must match the relay's voltage/current convention; the sector shown is not a universal setting.

Current and voltage coils

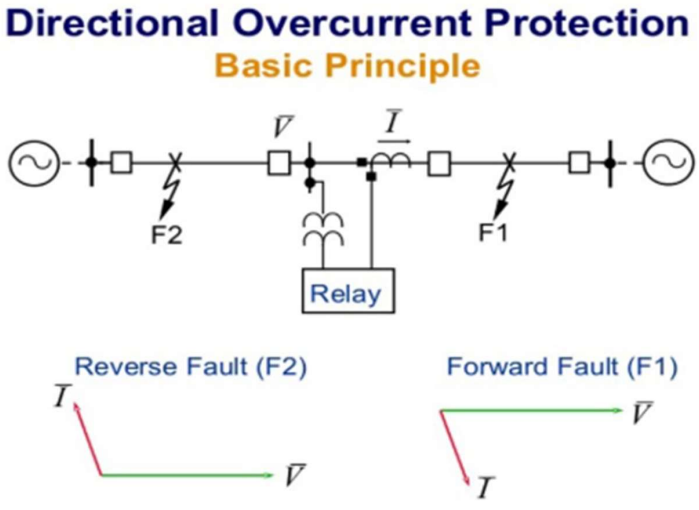
Class-slide anchor: Protection, p. 18.



A directional element combines current and a polarizing quantity. An ordinary nondirectional overcurrent element need not have a voltage coil.

Forward and reverse fault phasors

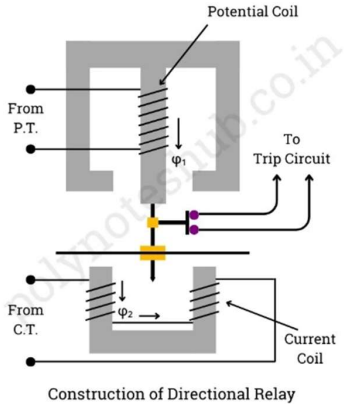
Class-slide anchor: Protection, p. 24.



Direction depends on the polarizing-voltage/current relationship and relay characteristic angle, not current magnitude alone.

Directional induction construction

Class-slide anchor: Protection, p. 25.



Construction of Directional Relay

Identify potential coil, current coil, moving disc and trip contacts. The configured torque direction defines forward operation; source prose must not be read as a universal reversed convention.

10.3 Distance: impedance as electrical reach

A distance element estimates *apparent impedance* from an appropriate voltage/current fault loop. For a simple fault on a uniform line, the measured loop impedance relates to impedance per length and distance to the fault. A low apparent impedance inside a selected characteristic can indicate a fault within reach.

For a phase-b-c loop,

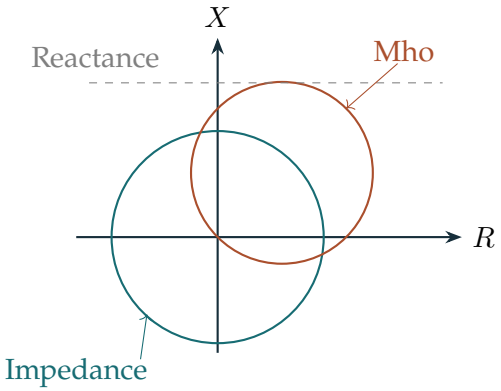
$$Z_{bc,app} = \frac{V_b - V_c}{I_b - I_c}.$$

For a compensated phase-a earth loop, one common convention is

$$Z_{aG,app} = \frac{V_a}{I_a + k_0(3I_0)}, \qquad k_0 = \frac{Z_{0,line} - Z_{1,line}}{3Z_{1,line}}.$$

Some relay texts absorb the factor three into a coefficient multiplying I_0 instead. Always match the current and coefficient definitions. This supplement shows why one bare V_a/I_a formula is not sufficient for every fault type.

Elementary characteristics in the R-X plane include **impedance** (circle centred at the origin), **reactance** (a reach boundary in reactance, requiring suitable directional/control logic) and **mho** (a circle through the origin along the characteristic angle, providing directional behaviour in its ideal form).



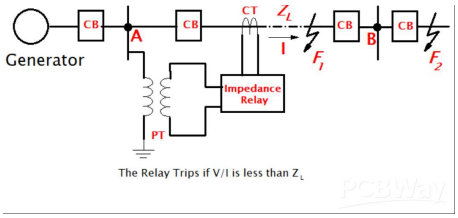
Separate ideal characteristic sketches superposed for comparison. Reach and polarisation in an actual relay require the specified implementation.

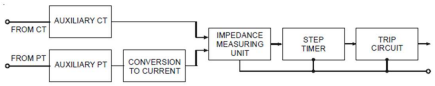
Zone 1 commonly underreaches the protected line and has no intentional delay. **Zone 2** extends beyond the remote end with a delay. **Zone 3** gives further delayed backup where required. The slides use 80%, 120%, approximately 0.35 s and 0.8 s as examples; they are not mandatory settings for every line. Reach must account for adjacent lines, measurement error, infeed and coordination.

Distance protection is less dependent on fault-current magnitude than simple OC, but it is not independent of all loading/source conditions. Fault resistance, remote infeed, power swings, load encroachment, CT/VT error and incorrect line data can change apparent impedance or the decision. A low $|V/I|$ must be assessed within its characteristic and supervision.

Distance sensing and processing

Class-slide anchor: Protection, p. 28.

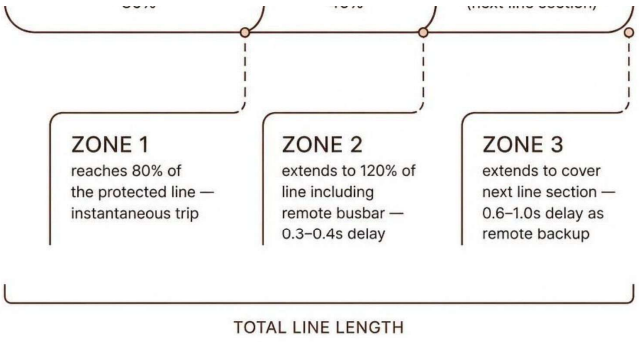




The selected voltage/current loop estimates apparent impedance. Fault resistance, infeed and instrument errors affect the estimate.

Distance reaches

Class-slide anchor: Protection, p. 30.



The class 80/120 percent settings and delays are illustrative. Zone reaches must coordinate with neighbouring lines.

Additional example: reach conversion to relay ohms

A line has $Z_L = 10 + \text{j}40\ \Omega$ primary. An 80% reach is $8 + \text{j}32\ \Omega$. CT ratio is $400/5 = 80$, VT ratio is $110000/110 = 1000$. Because $V_s = V_p/1000$ and $I_s = I_p/80$, relay impedance is $Z_s = Z_p(80/1000)$. Reach is therefore $0.64 + \text{j}2.56\ \Omega$ secondary. Interchanging the CT and VT ratios would give a severe reach error.

10.4 Differential: compare the zone's boundary currents

Apply KCL to currents crossing the protected equipment boundary. After CT ratio, polarity and any required transformer phase-shift compensation, external load/fault current ideally balances; an internal fault creates a nonzero sum.

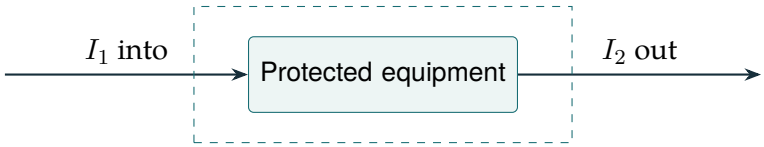
With I_1 defined into a two-terminal zone and I_2 defined out of it,

$$I_{diff} = |I_1 - I_2|, \qquad I_{bias} = \frac{|I_1| + |I_2|}{2}.$$

Alternatively define every current into the zone and use their algebraic sum. Do not mix these conventions. A simplified biased rule is

$$I_{diff} > I_{min} + sI_{bias},$$

where s is the restraint slope. Some devices instead use a maximum of pickup and slope thresholds, multiple slopes or another bias definition; specify the characteristic used.



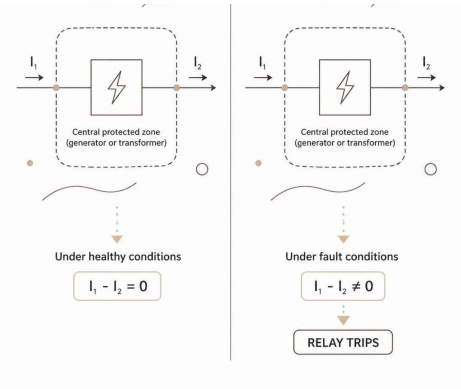
External through current: $I_1 \approx I_2$

Internal fault: compensated difference exceeds restraint

CT mismatch and saturation can create spill current during an external fault. Bias improves security while retaining internal-fault sensitivity. Transformer differential additionally needs ratio/vector-group compensation and suitable inrush/overexcitation restraint or blocking. Line differential may need communications and charging-current consideration. No scheme is automatically immune to every external condition.

Differential balance and internal fault

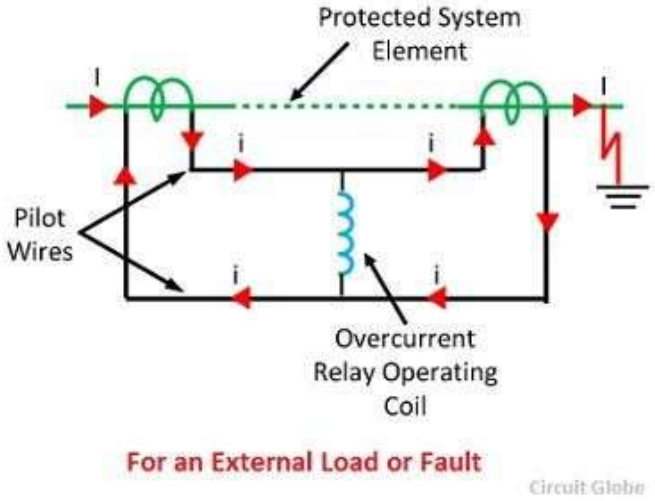
Class-slide anchor: Protection, p. 34.



With all currents referenced into the zone, their sum is approximately zero for through current and nonzero for an internal fault.

Pilot-wire differential circuit

Class-slide anchor: Protection, p. 35.



Trace CT secondary circulation during an external fault. Ideally no spill current flows through the operating coil.

Additional example: why bias is useful

Use the simplified rule with $I_{min} = 0.2$ A and $s = 0.30$. Measured compensated through currents $I_1 = 10$, $I_2 = 9.5$ A give $I_{diff} = 0.5$ A, $I_{bias} = 9.75$ A, threshold 3.125 A: restrained. For $I_1 = 6$, $I_2 = 1$ A, the difference is 5 A, bias 3.5 A, threshold 1.25 A: operates. These are educational values for the specified characteristic, not actual protection settings.

10.5 Compare the four principles

Principle	Main quantity	Typical use	Main limitation to remember
OC	Current magnitude and time	Radial feeders, backup	No inherent direction/location; coordination changes with fault level.
Directional	Angle relation / directional torque	Multi-source feeders, directional OC	Reference/polarity and polarisation matter.
Distance	Fault-loop apparent impedance	Transmission-line reach zones	Fault resistance, infeed, load/swing and measurement can affect reach.
Differential	Compensated current balance	Transformer, generator, bus, line	CT errors/saturation, compensation and communication where used.

11 Induction-type electromechanical relays

Class-slide anchor: Protection part pp. 18, 39–57.

The syllabus limits detailed electromechanical study to *induction type*. Attraction/solenoid relays appear in the slides as context and as instantaneous elements; their separate construction derivations are not needed here. Induction operation requires alternating flux and therefore does not operate from steady DC by the same principle.

Electromagnetic relay gallery

Class-slide anchor: Protection, p. 42.



These photographs provide recognition context. Attraction and induction are different mechanisms; the syllabus develops induction type.

11.1 Two fluxes and an aluminium moving element

Two alternating magnetic fluxes are displaced in *space* and *time*. Each induces eddy currents in a conducting nonmagnetic disc/cup. Interaction between one flux and currents induced by the other produces average torque. With sinusoidal fluxes

$$\phi_1 = \Phi_1 \sin \omega t, \qquad \phi_2 = \Phi_2 \sin(\omega t + \delta),$$

induced currents depend on flux rate of change. The net average cross-interaction is proportional to

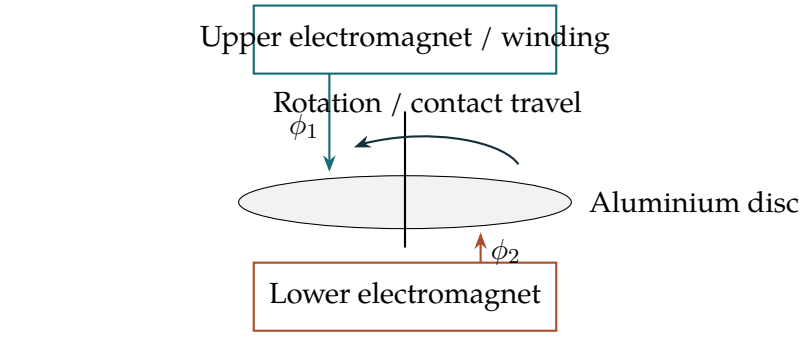
$$\boxed{T_{op} = K \Phi_1 \Phi_2 \sin \delta}.$$

If either flux is absent, or their phase displacement is zero, ideal net driving torque vanishes. For an OC construction with both fluxes proportional to the same relay current and approximately fixed phase displacement,

$$T_{net} \approx K_1 I^2 - K_2,$$

where K_2 represents spring/restraint torque in the simple model. Pickup occurs when driving torque exceeds restraint. A permanent magnet produces eddy-current braking approximately proportional to disc speed; bearings/friction and magnetic saturation modify the real curve.

It is *operating torque* that follows the approximate I^2 dependence. Disc speed is not universally proportional to I or exactly I^2: net torque, braking, saturation and acceleration matter. This corrects the inconsistent shorthand in the slide summaries.

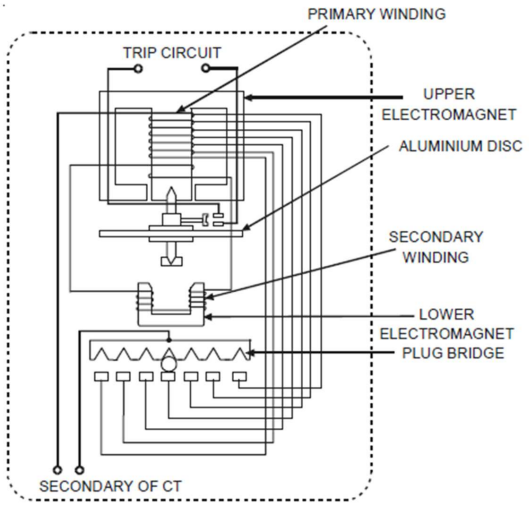


Spindle + spring restrain motion; braking magnet limits speed

Functional disc-relay sketch. Spatially displaced flux paths act on the disc; the sketch omits detailed iron-core and tap geometry.

Double-winding induction-disc construction

Class-slide anchor: Protection, p. 44.



Identify the two electromagnets, aluminium disc, spindle, spring, moving/fixed contacts and auxiliary winding connections.

11.2 Three ways to obtain the phase-displaced flux

Shaded pole: a copper shading ring surrounds part of a magnetic pole. Alternating main flux induces current in this ring, whose opposing field delays the shaded flux relative to the unshaded part. Both flux regions act on the disc to produce torque. The phase delay depends on the design; the slide's 40–50° is illustrative, not an invariant.

Watt-hour-meter / double-winding structure: an upper E-shaped magnet carries primary and secondary windings; an induced secondary current energises a lower U-shaped magnet. Circuit resistance/inductance produces a phase displacement between upper and lower fluxes. A tapped primary winding provides current-setting adjustment. Closing the designed secondary circuit enables its second flux and torque; opening that internal control path prevents the ideal driving torque. This is a description of the relay's magnetic circuit, not a suggestion to open an energised external CT secondary.

Induction cup: a light conducting cup rotates around a stationary iron core between two or more electromagnets. Phase-displaced fluxes create a rotating field and induced cup currents. Low inertia and high torque-to-inertia ratio permit fast response; it is used in suitable directional/distance constructions. The slide's approximately 0.01 s is an example, not every cup relay's operating time.

Structure	Source of displacement	Recognition point
Shaded pole	Shading-ring induced current	One pole face split into shaded/unshaded regions.
Double winding	Induced secondary and differing magnetic/electrical paths	Upper/lower magnets with disc between them.
Cup	Multiple phase-displaced coil fluxes	Light cup around fixed central core.

Shaded-pole induction structure

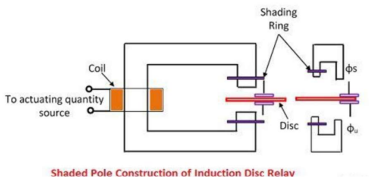
Class-slide anchor: Protection, p. 47.

a. Shaded pole structure

-**Coil Energization:**
- Utilizes current flowing through a single coil.
- Wound on a magnetic structure featuring an air gap.

Air-gap Fluxes:
- Produced by the initializing current.
- Split into two fluxes that displace in time-space due to a shaded ring.

Shaded Ring:
- Composed of a copper ring encircling part of each pole face.



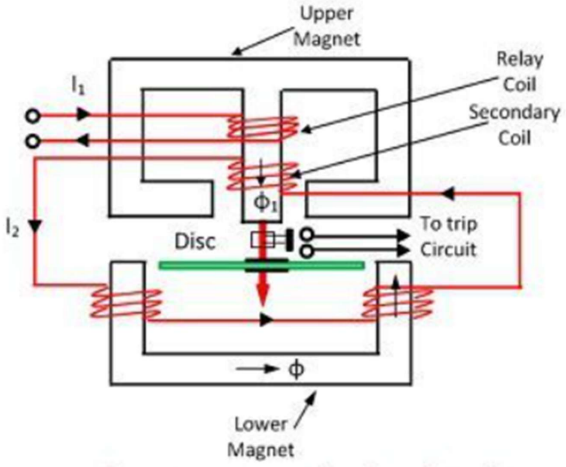
Aluminum Disc Characteristics:
- Made of aluminum with low inertia.
- Requires less deflecting torque for movement.

Induction Mechanism:
- Two rings have current induced by the alternating electromagnetic flux.
- Magnetic field from the current causes flux in the iron ring to lag phase by 40° to 50° behind the flux in the unshaded portion of the pole.

The copper shading ring delays one portion of the pole flux. Spatially and temporally displaced fluxes produce disc torque.

Watt-hour-meter induction structure

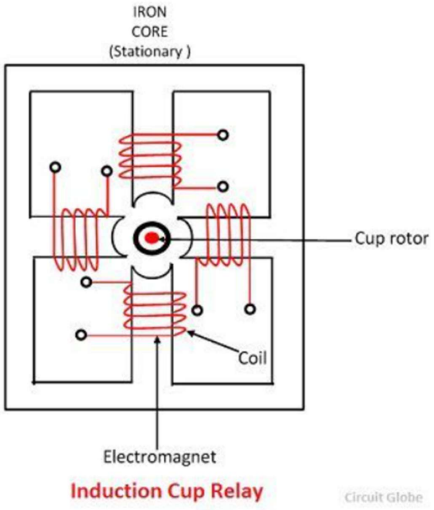
Class-slide anchor: Protection, p. 50.



The winding arrangement produces two phase-displaced fluxes through the disc. The permanent magnet supplies braking torque.

Induction-cup construction

Class-slide anchor: Protection, p. 53.



The light conducting cup rotates in the air gap around a stationary iron core. Multiple poles create the required flux interaction.

11.3 Pickup, time dial and inverse timing

Current tap / plug setting adjusts pickup. **Time dial** changes required disc travel/contact angle and thereby operating time at a stated current multiple. A larger travel generally delays operation. The curve shape is determined largely by the relay construction; moving a time dial does not turn every fixed mechanical curve into any desired curve.

Above pickup, higher current generally gives larger net torque and shorter time to travel to the contact. Magnetic saturation and design produce the chosen inverse/IDMT characteristic. "Inverse definite minimum time" means the curve approaches a minimum-time region; it does not mean delay is exactly $1/I$ at every current. After current falls, the spring returns the disc and contacts reset, with possible time/overshoot effects that matter in coordination.

Traditional separate phase elements were common, but the slide's requirement of a minimum three relay units is not universal: some three-phase three-wire OC arrangements can use two phase elements with appropriate ground protection. Describe the required protection coverage rather than assuming one specific device count.

Check your understanding

Why do two equal, in-phase fluxes fail to produce ideal induction torque?
Answer: $\sin\delta = 0$ when their phase displacement is zero. The cross interactions cancel on average. The moving element needs both spatial and time displacement of the AC fluxes.

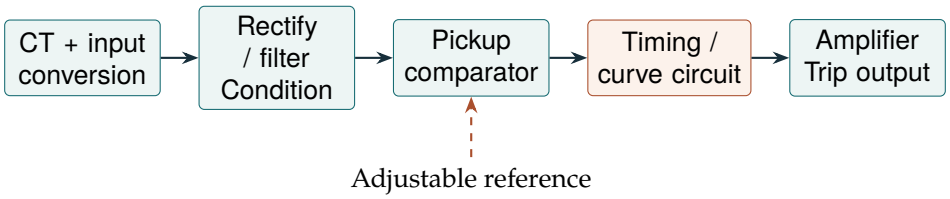
12 Static, microprocessor and numerical relays

Class-slide anchor: Protection part pp. 12, 58–74.

12.1 Static relays: the decision is electronic

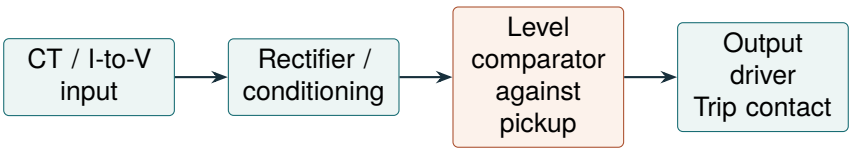
Static relays use analogue electronic measuring/comparing circuits rather than a moving disc for the decision. They can still have electromechanical output contacts; "static" does not imply every part of the trip chain has no movement. Auxiliary power supplies, transducer isolation, component drift and surge immunity are relevant.

Static time-overcurrent block diagram



The CT-derived signal is converted to a usable voltage and rectified/filtered or otherwise conditioned. A comparator recognises pickup. A timing circuit, such as an RC charging arrangement with current-dependent charging/threshold control, implements the required delay. An amplifier/output device energises the trip stage. Functional ordering can differ: the class diagram places a level comparator after the timing stage. Either drawing must explain where current pickup and timer-completion decisions occur. Auxiliary DC supply supports the electronic and output circuits.

Static instantaneous overcurrent block diagram



There is no intentional timing network in the 50 path. Filtering, comparator response and output contacts still take time. The slide's 8–16 ms is an illustrative operating range, not the definition of instantaneous. If directional supervision is added, a VT path and phase comparator supply permission to the OC output/timer; this covers the directional static diagram in the source.

Static device gallery

Class-slide anchor: Protection, p. 59.



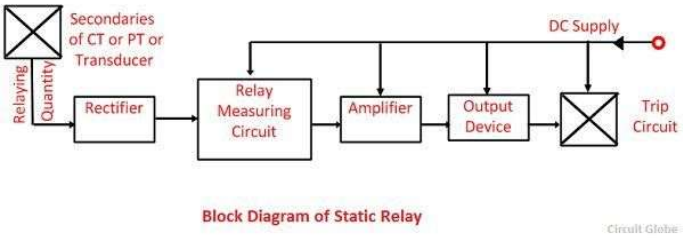




Electronic measuring circuits replace the moving measuring element. The output may still use an electromechanical auxiliary contact.

Static and directional static blocks

Class-slide anchor: Protection, p. 61.



Block Diagram of Static Relay

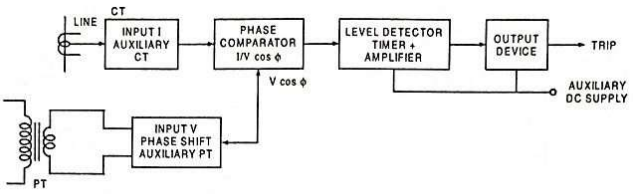


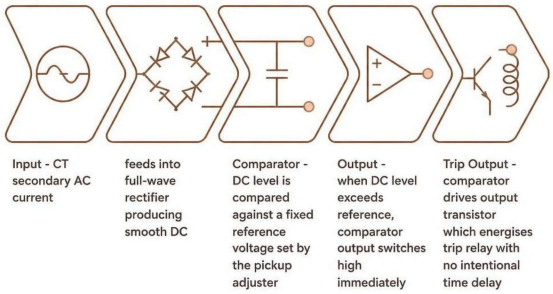
Fig. 10.39. *Block Diagram of Directional Static Overcurrent Relay*

The first diagram shows rectifier, measuring circuit, amplifier, output and trip supply. The second includes phase comparison for direction. Compare the clearly separated time-overcurrent and instantaneous diagrams in Section **??**.

Instantaneous static chain

Class-slide anchor: Protection, p. 62.

Static Instantaneous (50): Block Diagram



Instantaneous Element Logic

The static 50 element is considerably simpler than the 51 — it contains no timing network. The key signal path is:

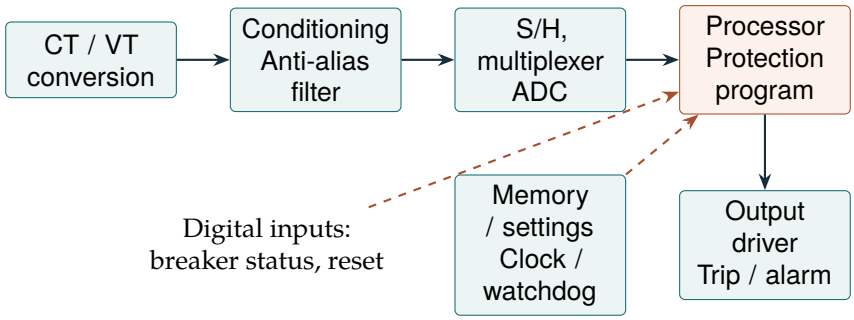
Current → Rectifier → Level Comparator → Instantaneous Trip Output

- No RC timing circuit — comparison is direct and immediate
- Pickup threshold set by an adjustable reference voltage
- Operating time is typically **8–16 ms** (half to one cycle)
- Output transistor drives an interposing output relay to trip the breaker

Rectification/conditioning, comparison and output trigger replace the induction delay. A comparator reference determines pickup.

12.2 Microprocessor relay: sampling and a programmed decision

A hardware architecture includes CT/VT input conversion/isolation, signal conditioning, anti-alias filtering, sample-and-hold where required, analogue multiplexer for multiple channels if used, ADC, processor, memory, clock, digital inputs, output drivers and an auxiliary supply.



The class's older CT → I/V → rectifier → multiplexer → ADC → 8085 arrangement is a particular magnitude-based implementation. A modern waveform-sampling numerical relay can digitise conditioned AC directly; a rectifier is not mandatory before every ADC. ADC, multiplexer and processor ports handle measurement/control data, not power-line fault current.

Digital relay recognition

Class-slide anchor: Protection, p. 64.



Front-panel controls and indication are interfaces; the protection computation occurs inside the device.

Basic microcomputer protection blocks

Class-slide anchor: Protection, p. 65.

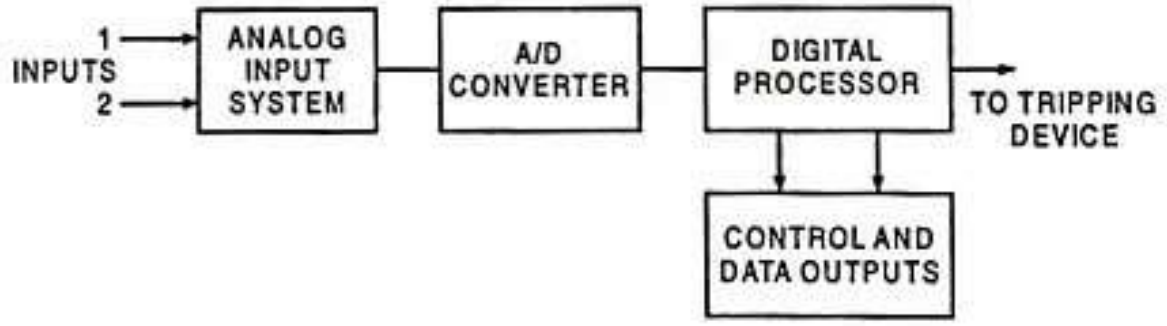
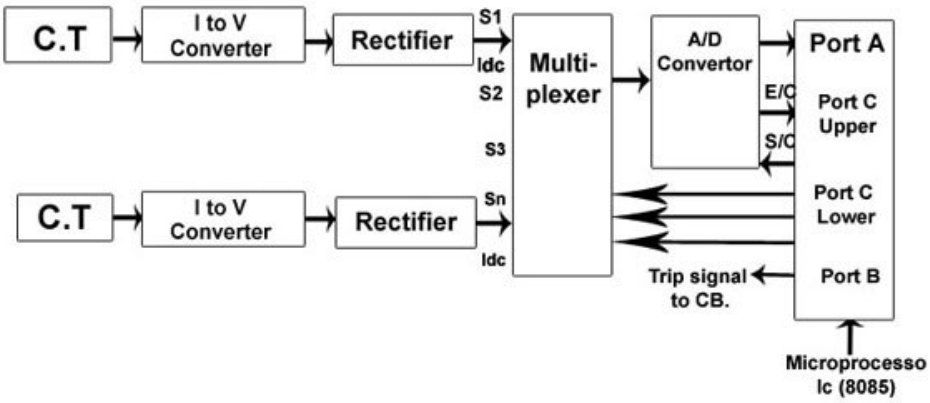


Fig. 10.58. ***Basic Protection Scheme Using Microcomputer***

Analogue input conditioning precedes A/D conversion; the digital processor performs the protection decision.

Historical 8085 overcurrent hardware

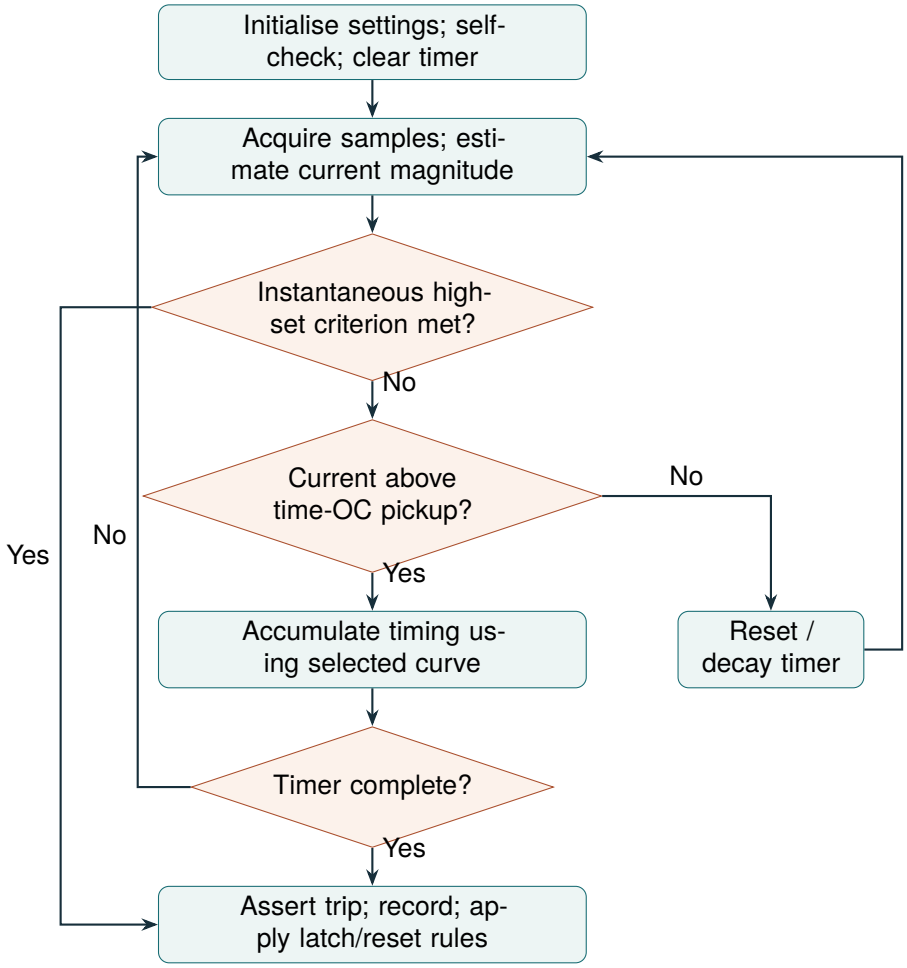
Class-slide anchor: Protection, p. 67.



Block diagram of a microprocessor-based overcurrent relay

This class design rectifies CT-derived voltages before multiplexing and A/D conversion. It estimates an overcurrent magnitude; do not claim this rectified waveform retains AC phase for a general phasor relay.

12.3 Overcurrent flowchart and timing accumulator



Educational 50/51 decision flow. Repeated acquisition is continuous; reset, supervision and latching follow the specified relay algorithm.

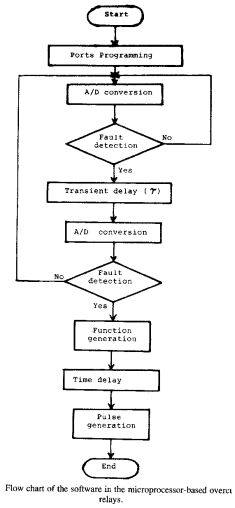
For a constant overcurrent, a definite-time implementation counts elapsed samples. For changing current with an inverse curve, one useful educational accumulator is

$$q \leftarrow q + \frac{\Delta t}{t(M)}, \qquad \text{trip when } q \geq 1.$$

This models progress toward operation rather than restarting a full timer every sample. Below pickup, the chosen reset rule clears or decays q. The source flowchart uses an initial threshold/delay, repeated checking, time setting and execution of trip; the more explicit chart above makes the continuing pickup and reset decisions visible.

Original microprocessor flowchart

Class-slide anchor: Protection, p. 66.



Flow chart of the software in the microprocessor-based overcu relays.

Follow acquisition, pickup decision, delay, repeated current check, trip and reset. The main text adds explicit dropout handling and a clearer loop.

12.4 Fundamentals of a numerical relay

A numerical relay implements measurements and protection algorithms on digitally represented samples, often with a microprocessor/DSP. “Microprocessor” and “numerical” overlap in practice: an early processor-based relay may make simple magnitude/time decisions; numerical systems commonly provide sampled-waveform estimation and many functions. They are not completely separate physical principles.

An anti-alias filter limits high-frequency components before sampling. The Nyquist condition alone is not a complete relay design; filter response, sampling rate, estimator window and fault transients matter. Fundamental-frequency phasors may be estimated with a DFT or other algorithms. ADC range/resolution, CT saturation, frequency deviation, DC offset and harmonics influence measurement; filtering is not a guarantee to remove every error.

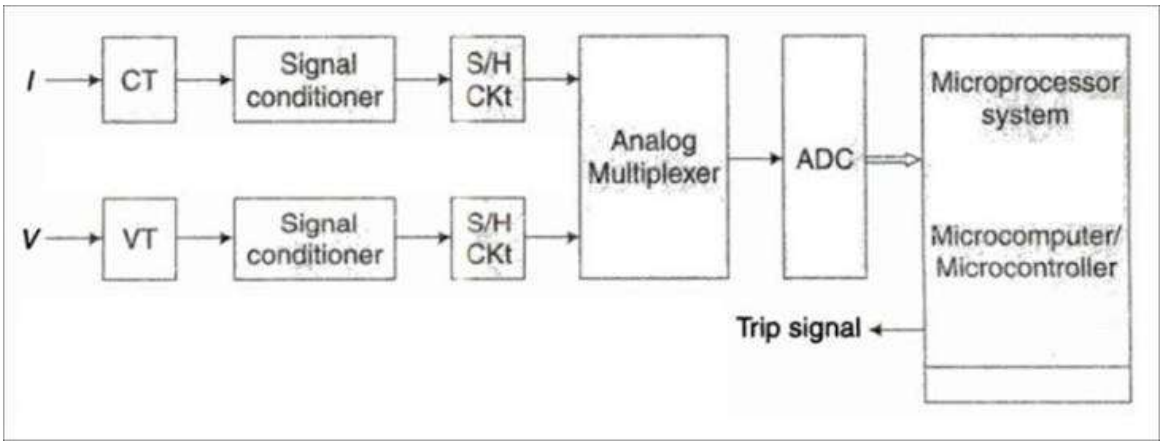
Typical functions include programmable OC/directional/distance/differential elements, event and waveform recording, metering, setting groups, self-diagnostics, breaker monitoring, communications and logic. Advantages include repeatable characteristics, multifunction capability and easier records/settings management. Limitations include dependence on auxiliary power and correct firmware/settings, possible common hardware/software failures, measurement transients and the need for secure access and testing. Self-diagnostics detect some faults, not every failure.

The slides’ 16–64 samples/cycle, 0.01-step settings, specific chassis sizes and proprietary logic examples describe particular implementations, not all numerical relays. IEC 61850 provides substation

communication/data-model capabilities; using it does not automatically guarantee speed, redundancy or cybersecurity. Adaptive/AI-based functions are contextual material, not a replacement for understanding the deterministic protection conditions in this module.

Sample-and-hold numerical hardware

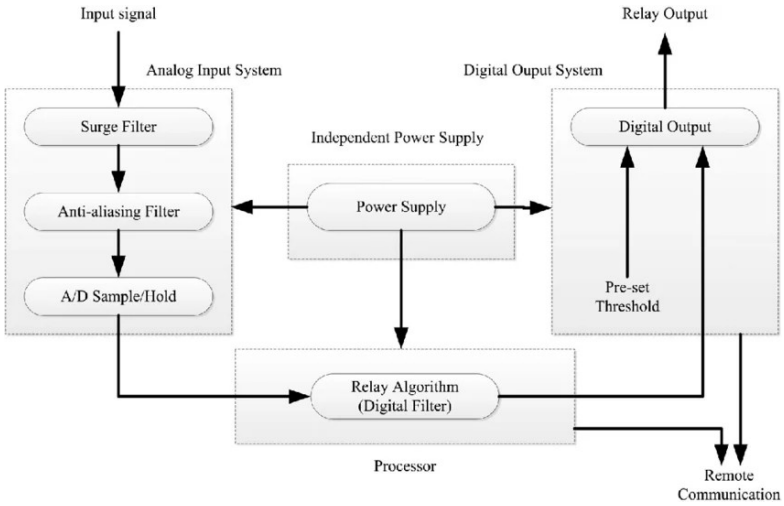
Class-slide anchor: Protection, p. 72.



CT/VT signals are conditioned, sampled, multiplexed, converted and processed. Anti-alias filtering is explained in the main text.

Numerical relay signal and power paths

Class-slide anchor: Protection, p. 73.



Surge filtering and anti-alias filtering protect the analogue input; the processor runs protection logic and sends digital outputs and communications.

Feature	Induction electromechanical	Static analogue	Numerical
Decision	Flux, torque, moving disc/cup	Electronic comparator/timer	Sampled values + software
Timing settings	Tap, dial; construction curve	Electronic references/RC circuit	Pickup, curve, TMS and reset logic
Records / functions	Usually limited	Usually limited by circuitry	Commonly multiple functions and stored records
What can still move?	Disc/cup and contacts	Output relay / breaker	Output contacts / breaker
Key checks	Mechanical condition, pickup/timing	Supply, drift, surge immunity	Supply, settings, firmware, input/logic performance

13 Circuit breakers and the arc-extinction problem

Class-slide anchor: Syllabus supplement: Module 3 circuit-breaker items on syllabus PDF p. 2. The supplied protection deck provides relay/trip context but no dedicated breaker section..

13.1 What the breaker must accomplish

A circuit breaker makes, carries and interrupts current within its assigned ratings, including specified fault duties. Its closed contacts offer a low-resistance conducting path. A trip command releases an operating mechanism so contacts separate; successful interruption requires the conducting arc to die and the contact gap to remain insulating under the voltage that returns across it.

The trip coil generally releases stored operating energy; the coil need not itself supply all contact-separation energy. Breaker *opening time* runs to contact separation; *arcing time* runs from arc initiation to final extinction; their sum after the applicable initiation defines interruption time according to the stated convention. Fault clearing also includes preceding relay delays. These intervals should not be counted twice.

An isolator/disconnector establishes isolation under its specified switching duties; it is not a substitute for a fault-interrupting circuit breaker. A fuse interrupts by melting its element and extinguishing its arc, and normally must be replaced after operation. The breaker can be operated again after the fault and its condition/duty permit it.

13.2 Why opening contacts initially produces an arc

At separation the last contact area becomes small, current density and local temperature increase, and a conducting plasma path can form. Gas ionisation and/or vaporised contact material sustain the arc depending on the medium. Current continues after the metal contacts have separated. Inductance resists abrupt current change; the circuit can produce a high voltage if interruption attempts a rapid di/dt:

$$v_L = L\frac{di}{dt}, \qquad W_L = \tfrac{1}{2}Li^2.$$

The arc converts electrical energy to heat. Its voltage, current, contact geometry and cooling determine its behaviour; it is not simply a fixed resistor. A wider gap alone does not prove that the arc will extinguish or remain extinguished.

13.3 The two elementary arc-extinction approaches

High-resistance method: increase the arc's effective resistance/voltage enough to force its current down. Cooling, lengthening, narrowing/constriction and splitting into multiple series arcs are familiar methods. Arc chutes and magnetic movement can assist suitable devices. The method dissipates energy and can produce switching overvoltage; application depends on the switch design and circuit duty, and is associated with many lower-voltage/current-limiting and DC interruption arrangements.

Low-resistance / current-zero method: in an AC breaker, allow the arc to conduct until a natural current zero, then rapidly restore insulation so current does not restart. The arc need not be forced to a very high resistance throughout the preceding half-cycle. Cooling, removal of hot ionised material, recombination, electron attachment in suitable gases, or rapid loss/condensation of conducting vapour help the post-zero gap recover.

The term "low resistance" describes the conducting arc before extinction; it does not mean the open breaker should have a low resistance. Both methods manage heat, charge carriers and recovery voltage. AC natural-current-zero interruption is central to the SF_6 and VCB explanations below. DC has no periodic natural current zero and needs a suitable dedicated interruption/current-zero-creation mechanism; an AC rating cannot simply be reused for DC.

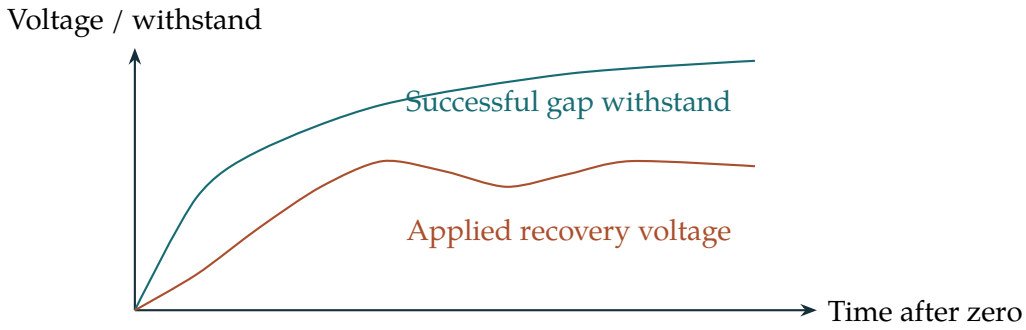
13.4 Current zero, recovery voltage and the race for insulation

Near AC current zero, energy input to the arc is reduced and the conductive channel can collapse. The network then applies a voltage across the separating contacts. The immediate transient is **transient recovery voltage** (**TRV**); after transients subside, the power-frequency voltage across the open gap is the recovery voltage. Traditional texts often use "restriking voltage" for the initial transient. Actual reignition/restrike means conduction restarts because interruption failed at that stage.

Success requires both:

- **Thermal recovery:** residual post-arc current and remaining heating must not rebuild the hot conducting path immediately after zero.
- **Dielectric recovery:** the recovering gap's withstand capability must remain above the applied recovery-voltage stress as time progresses.

A common introductory description compares rate of rise of recovery voltage with rate of dielectric recovery; the full voltage-versus-time curves and thermal behaviour matter, not just one slope. TRV depends on circuit inductance, capacitance, source/fault location and switching conditions.



Conceptual successful interruption: withstand exceeds applied stress. Not a measured or standardised TRV test waveform; near-zero thermal recovery must also succeed.

13.5 An optional elementary LC recovery-voltage derivation

This derivation helps explain the source of a transient; it is not an additional numerical requirement in the supplied syllabus. Consider a simplified lossless source-side circuit after current zero: approximately constant source voltage E_m over the short transient, inductance L and effective capacitance C across the open break. With $v_C(0) = 0$, $i(0) = 0$,

$$L\frac{di}{dt} + v_C = E_m, \quad i = C\frac{dv_C}{dt} \quad \Rightarrow \quad LC\frac{d^2v_C}{dt^2} + v_C = E_m.$$

The solution is

$$v_C(t) = E_m[1 - \cos(\omega_0 t)], \qquad \omega_0 = 1/\sqrt{LC}.$$

The first ideal peak is $2E_m$ at $t = \pi/\omega_0$. Average rise to this peak is $2E_m\omega_0/\pi$; maximum instantaneous slope is $E_m\omega_0$. The initial derivative in this special model is zero, so the average rise is not the derivative at $t = 0$. Real damping and distributed network dynamics change the waveform and peaks; do not treat $2E_m$ as every breaker TRV.

13.6 Current chopping and reclosing context

Current chopping interrupts a small inductive current before its natural zero. The stored inductive energy can charge capacitance and cause overvoltage. An ideal energy estimate is

$$\tfrac{1}{2}LI_{ch}^2 = \tfrac{1}{2}C(\Delta V)^2 \quad \Rightarrow \quad \Delta V \approx I_{ch}\sqrt{L/C},$$

under the deliberately simplified zero-initial-capacitor-voltage, lossless model. Contact material, design and suitable surge protection reduce troublesome effects. Chopping, restrike and normal current-zero interruption are different concepts.

Automatic reclosing may restore supply after a transient line fault; permanent faults require lockout according to the scheme. A breaker must withstand its specified operating duty, fault making, interruption and repeated operations. Relay reset/trip-latch logic and breaker readiness therefore belong to the overall protection chain.

13.7 Classification and key ratings

By arc-extinction medium, elementary classifications include oil (bulk/minimum oil), air/air blast, SF_6/other gas, and vacuum. By construction, live-tank and dead-tank arrangements distinguish whether the interrupter enclosure is at line potential or earth potential. These are different classification axes. The syllabus asks detailed study of SF_6 and vacuum.

Key ratings include rated voltage and insulation withstand, continuous current, short-circuit breaking current, peak making current, short-time withstand current and specified operating duty/TRV capability. Breaking current is commonly stated as an RMS symmetrical component with specified conditions; peak making duty accounts for asymmetry. The old three-phase breaking-capacity expression

$$S_{break}(\text{MVA}) = \sqrt{3}V_{rated}(\text{kV})I_{break}(\text{kA})$$

expresses a duty convention, not actual useful power delivered into a bolted fault. Practical selection requires all applicable ratings rather than MVA alone.

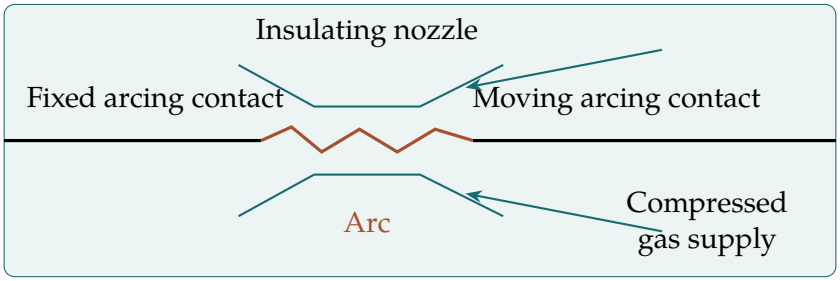
14 SF6 and vacuum circuit breakers

Class-slide anchor: Syllabus supplement; primary technical references are linked in Section ??..

14.1 SF6: gas insulation and current-zero interruption

Sulphur hexafluoride (SF_6) provides strong insulation and useful arc-interruption properties. It is electronegative: capture of free electrons helps reduce conducting charge carriers in the recovering gas. Cooling and replacement/removal of hot arc gas are also essential; electron attachment alone is not the complete interruption mechanism.

An elementary **puffer** interrupter contains fixed/moving main contacts, arcing contacts, an insulating nozzle and a gas compression chamber/piston associated with the operating mechanism. The opening movement compresses gas and directs it through the arcing region. Main contacts carry normal current and separate before the arcing contacts; the latter carry the arc. Self-blast/thermal-assisted designs use arc energy to assist gas pressure and reduce mechanism energy for suitable duties.



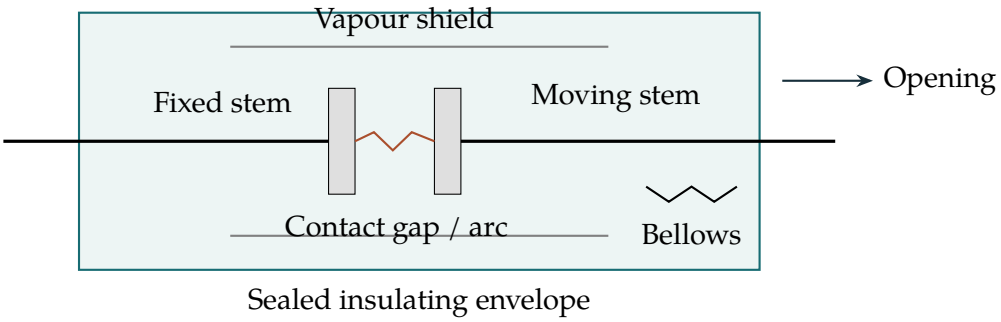
Functional puffer gas-flow sketch, not a detailed engineering cross-section. The separate main-contact path and compression piston are omitted for clarity.

Opening sequence: trip releases the mechanism; main contacts separate and current transfers to the arcing path; arcing contacts part and an arc forms; gas compression/nozzle flow cools the channel and carries hot products away; at current zero the gap recovers; continuing contact travel and cool insulating gas allow it to withstand TRV. Reignition is possible if gas flow or recovery is inadequate. This sequence is cross-checked against Schneider Electric's puffer explanation and ABB's breaker description.

Advantages include effective interruption/insulation and compact arrangements for suitable duties, without oil as the interrupting medium. Limitations include sealed-gas-system requirements, pressure/density monitoring, leakage management and handling of arc decomposition products. SF_6 is a strong greenhouse gas; containment, recovery and reduced-emission alternatives matter. No specific regulatory rule or future phase-out date is asserted here.

14.2 VCB: a metal-vapour arc in a sealed vacuum interrupter

A vacuum circuit breaker uses a hermetically sealed interrupter with fixed and moving contacts, insulating envelope (often ceramic), metal end seals, a flexible metal bellows for moving-contact travel and an internal vapour shield. The vacuum is the interrupting environment; the whole switchgear assembly still requires external insulation.



Labelled conceptual vacuum bottle. Bellows representation is schematic; it preserves a vacuum seal while allowing axial movement.

Vacuum does *not* mean no arc can form. Contact separation produces metal vapour from the electrodes, which supplies the conducting plasma. Suitable contact geometry can use magnetic fields to distribute arc energy and avoid local overheating. Near current zero, the vapour arc can no longer be sustained; vapour rapidly condenses on internal metal surfaces and the contact gap recovers dielectric strength. This mechanism is cross-checked against Siemens' vacuum-interrupter technical descriptions.

Advantages include a short required contact gap, rapid dielectric recovery, low arc energy and a sealed interrupting unit without an SF_6 gas-interruption system. Vacuum interrupters are widely used in medium-voltage applications and suitable specialised duties; there is no universal maximum voltage to memorise independently of device design.

Limitations/considerations include vacuum integrity, contact erosion and switching duty, current-chopping/restrike overvoltages for some inductive loads, and suitable protection/insulation coordination. A sealed interrupter may require replacement rather than internal field repair. The mechanism, connections and external insulation still need maintenance/checks even when the vacuum bottle is described as maintenance-free.

14.3 Comparison that can be written in an exam

Aspect	SF_6 breaker	Vacuum breaker
Interrupting environment	Controlled insulating gas	Sealed evacuated chamber
Arc medium	Hot ionised gas with electrode products	Primarily contact-metal vapour
Extinction emphasis	Gas flow cools the arc; insulation recovers at current zero	Vapour plasma collapses/condenses; gap recovers at current zero
Core construction	Arcing/main contacts, nozzle, compression chamber	Contacts, sealed envelope, bellows, vapour shield
System considerations	Gas density, leakage and decomposition products; greenhouse impact	Vacuum seal/contact condition; inductive-load switching overvoltage
External insulation	External insulation still required	External insulation still required

Check your understanding

What must happen after the arc reaches current zero?
Answer: The conducting channel must lose conductivity, and the gap must recover enough dielectric strength to withstand the returning voltage. Current zero alone does not guarantee interruption.

15 Additional integrated practice

15.1 From nameplate reactance to bus fault current

A 25 MVA, 11 kV generator with $X''_d = 0.20$ p.u. feeds an 11/66 kV transformer rated 25 MVA with $X_T = 0.10$ p.u.; a line has $X_L = 20\ \Omega$/phase on the 66 kV side. Find solid three-phase fault current at the far end, neglecting load/resistance and assuming $E = 1$. Select system base 50 MVA, 11/66 kV.

$$X_G = 0.20(50/25) = 0.40, \quad X_T = 0.10(50/25) = 0.20, \quad Z_{B,66} = 66^2/50 = 87.12\ \Omega,$$

$$X_L = 20/87.12 = 0.2295684, \quad X_{1,eq} = 0.8295684, \quad |I_f| = 1/0.8295684 = 1.205446 \text{ p.u.}$$

$$I_{B,66} = 50/(\sqrt{3}\,66) = 0.4373866 \text{ kA}, \quad \boxed{|I_f| = 0.527246 \text{ kA}}.$$

Short-circuit level is $50/0.8295684 = 60.2723$ MVA. The answer is far-end current at 66 kV; multiplying by the 11 kV base current instead would give the referred generator-side current, a different quantity. No additional ideal-ratio impedance appears in the p.u. series sum.

15.2 Finite ground-fault resistance changes both magnitude and angle

At a bus, $E = 1$, $Z_1 = \text{j}0.2$, $Z_2 = \text{j}0.2$, $Z_0 = \text{j}0.1$ p.u. An a-ground fault has *resistance* $Z_f = 0.05$ p.u. Then

$$I_1 = \frac{1}{0.15 + \text{j}0.5} = 0.550459 - \text{j}1.834862,$$

$$\boxed{I_a = 1.651376 - \text{j}5.504587 \text{ p.u.}}, \quad |I_a| = 5.746958 \text{ p.u.}, \quad \angle I_a \approx -73.3008^\circ.$$

$V_a = Z_f I_a = 0.0825688 - \text{j}0.2752294$ p.u. All three sequence currents are equal; the unfaulted fault-branch currents remain zero. Resistance cannot be inserted as another imaginary reactance.

15.3 A relay-trip decision with a specified reset rule

For a definite-time OC element, pickup is 400 A primary and delay is 0.30 s. Current is 600 A for 0.18 s, then 350 A for 0.05 s, then 600 A for 0.20 s. With *instantaneous timer reset* below pickup, neither above-pickup interval reaches 0.30 s; there is no delayed trip. If the problem instead specifies a slow-decay reset, the answer can change. This is why a numerical relay's reset setting belongs in the coordination model.

16 Revision sheets and error-proof workflows

16.1 Base quantities and sequence transformation

Keep these conventions on every calculation page

Three-phase base uses *total* S_B and *line-to-line* V_B:

$$I_B = \frac{S_B}{\sqrt{3}V_B}, \quad Z_B = \frac{V_B^2}{S_B}, \quad Z_{\text{pu,new}} = Z_{\text{pu,old}} \frac{S_{B,new}}{S_{B,old}} \left(\frac{V_{B,old}}{V_{B,new}} \right)^2.$$

Keep S_B common throughout the network and choose voltage bases across each transformer in its rated line-voltage ratio. Each zone has its own I_B and Z_B.

$$a = -\frac{1}{2} + \mathrm{j}\frac{\sqrt{3}}{2}, \quad a^2 = -\frac{1}{2} - \mathrm{j}\frac{\sqrt{3}}{2}, \quad a^3 = 1, \quad 1 + a + a^2 = 0.$$

$$\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix}, \qquad \begin{bmatrix} I_a \\ I_b \\ I_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix}.$$

Replace I by V for voltages. The factor $1/3$ belongs to the phase-to-sequence transformation. $I_a + I_b + I_c = 3I_0$ is the residual/earth-return current under the chosen reference direction.

16.2 Fault equations on one page

Use common-base Thevenin impedances at the fault bus. In every row,

$$V_1 = E - Z_1 I_1, \qquad V_2 = -Z_2 I_2, \qquad V_0 = -Z_0 I_0.$$

Fault	Sequence-current equation	Phase condition / connection
3-phase solid	$I_1 = E/Z_1$, $I_0 = I_2 = 0$	Positive network only; $I_a = I_1$, $I_b = a^2 I_1$, $I_c = aI_1$.
a-ground	$I_0 = I_1 = I_2 = E/(Z_1 + Z_2 + Z_0 + 3Z_f)$	Three sequence networks in series; $I_a = 3I_1$; $I_b = I_c = 0$; $V_a = Z_f I_a$.
b-c	$I_1 = E/(Z_1 + Z_2 + Z_f)$, $I_2 = -I_1$, $I_0 = 0$	$I_a = 0$, $I_b = -\mathrm{j}\sqrt{3}I_1$, $I_c = -I_b$; Z_f is impedance between the faulted phases.
b-c-ground	$I_1 = \dfrac{E}{Z_1 + \frac{Z_2(Z_0+3Z_f)}{Z_2+Z_0+3Z_f}}$	Positive network feeding negative and modified zero networks in parallel; Z_f is the common path to ground.
	$I_2 = -I_1 \dfrac{Z_0 + 3Z_f}{Z_2 + Z_0 + 3Z_f}$	$I_a = 0$; $V_b = V_c = Z_f(I_b + I_c)$.
	$I_0 = -I_1 \dfrac{Z_2}{Z_2 + Z_0 + 3Z_f}$	Reconstruct I_b, I_c using the full inverse transformation.

For the solid LL case, $V_1 = V_2$ and $V_0 = 0$. For the solid LLG case, $V_0 = V_1 = V_2$; these equal values are *sequence terminal voltages*, not three phase voltages. For LG with a disconnected zero-sequence network, the ideal steady-state sequence model gives no metallic earth-fault path; capacitances may still permit current in a more complete model.

16.3 A six-step fault-analysis routine

1. Draw the one-line network, mark the faulted bus and identify connections/grounding. State fault type, phase labels, fault impedance and whether load is neglected.
2. Select a common power base and compatible voltage bases. Convert every impedance before combining it. Include $3Z_n$ in the zero-sequence neutral path.
3. Draw positive, negative and zero networks, retaining only physically available zero-sequence paths. Reduce each to the driving-point Thevenin equivalent at the fault bus.
4. Apply phase boundary conditions to connect the sequence networks. Calculate I_0, I_1, I_2 as *complex* quantities, including their signs.
5. Reconstruct phase currents and, if requested, sequence and phase voltages. Convert to amperes with the current base *at the fault location*.
6. Check the unfaulted fault-branch currents, equal/opposite LL currents, earth-current sum, and voltage conditions. Round only the final result.

Checks that catch most mistakes

$1/(\mathrm{j}X) = -\mathrm{j}/X$, but $-V/(\mathrm{j}X) = +\mathrm{j}V/X$. A sign error in the second expression corrupts an LLG reconstruction. In a b-c-ground fault, I_a must be zero in the fault branch; $3I_0$ is the total earth current, not automatically one faulted-phase current. A computed $I_a \neq 0$ is a failed boundary condition, even if another magnitude looks plausible.

16.4 Protection decisions to recognise immediately

Principle	Decision quantity	Essential check
Overcurrent	$M = I/I_{pickup}$; trip after selected time if $M > 1$	Load allowance, minimum fault sensitivity, downstream coordination.
Directional OC	Directional voltage/current comparison enables OC	Reference polarity and polarising signal during the relevant fault.
Distance	Compensated fault-loop $Z_{app} = V_{loop}/I_{loop}$	Correct loop, CT/VT conversion, reach, infeed and fault resistance.
Differential	$I_{op} = \lvert I_1 - I_2 \rvert$ after ratio/polarity compensation	External-fault stability; CT saturation, bias, transformer inrush.

For the standard inverse example used here,

$$t = TMS\frac{0.14}{M^{0.02} - 1} \quad (M > 1).$$

Curve family, time multiplier, pickup, reset rule and instantaneous setting all matter. Primary and backup protection are *roles*; induction, static and numerical are *technologies*; OC/distance/differential are *principles*. Do not mix these classifications.

For a circuit breaker, the relay orders a trip, the mechanism separates contacts, an arc carries current temporarily, and the interrupting medium must remove conductivity and withstand the returning voltage. In an AC breaker, the decisive interval lies around current zero. SF_6 uses controlled gas action; a VCB extinguishes a contact-metal-vapour arc inside an evacuated interrupter.

16.5 Common confusions: make the distinction explicit

Tempting shortcut	Correct interpretation
Use line voltage with per-phase VA in V^2/S	Use line voltage with total three-phase VA, or phase voltage with per-phase VA.
All equipment reactances can be added as given	Nameplate p.u. values must first use a common base.
A delta blocks every zero-sequence effect	It blocks external zero-sequence line current on that side; internal circulation can support a grounded-wye-side path.
Negative sequence has a separate generator EMF	The balanced pre-fault source exists in the positive network; negative and zero networks are passive in this model.
Fault always means the largest possible current	High-impedance faults and weak infeed can produce limited current; three-phase need not be the largest fault.
An instantaneous element takes zero seconds	It has no intentional delay; measuring, output and breaker times remain.
An induction relay works from one alternating flux alone	Net average induction torque needs two spatially and temporally displaced fluxes.
A microprocessor OC relay is a complete modern numerical relay	It is one implementation/function; modern multifunction numerical relays extend measurement, logic and supervision.
An isolator can interrupt the short circuit	A conventional isolator provides isolation after interruption; fault interruption requires a suitably rated breaker.
No air in a VCB means no arc	Contact-metal vapour forms the arc during interruption.

17 Exam practice with answers

17.1 How to use the supplied examination pattern

The syllabus specifies a 60-mark, 2 hour 30 minute end-semester examination. Part A has eight compulsory 3-mark questions, two from each module, totalling 24 marks. Part B offers two 9-mark full questions from each module; answer one from each pair, totalling 36 marks. A full question has at most three subparts. The prompts below follow those sizes for practice; they are not claimed predictions or past-paper questions.

For a numerical answer, give the base, equivalent network, formula, complex substitution and units. For a descriptive answer, start with the operating quantity, draw and label the diagram, explain the sequence of operation, and mention the important limitation. A labelled diagram plus a connected explanation earns more confidence than an isolated list of component names.

17.2 Three-mark self-test: attempt before reading the answer

Check your understanding

1. Derive three-phase impedance base using line quantities.
Answer: Using $V_{B,ph} = V_{B,LL}/\sqrt{3}$ and $S_{B,ph} = S_B/3$, $Z_B = V_{B,ph}^2/S_{B,ph} = V_{B,LL}^2/S_B$. The current base remains $S_B/(\sqrt{3}V_{B,LL})$.

Check your understanding

2. Give two benefits and one condition of the per-unit method.
Answer: It simplifies transformer referral and puts differently rated equipment on comparable normalised scales. The condition is a common system power base with transformer-compatible voltage bases; convert old apparatus bases first.

Check your understanding

3. Define positive, negative and zero sequence.
Answer: Positive sequence is an equal-magnitude balanced set in the chosen a-b-c order; negative sequence has the reverse order. Zero-sequence phasors are equal in magnitude and phase, so their sum is three times one component.

Check your understanding

4. Why does neutral impedance enter as $3Z_n$?
Answer: Each zero-sequence phase current equals I_0; the neutral carries $3I_0$. Its voltage drop $3I_0Z_n$ contributes an equivalent $3Z_n$ in the per-phase zero network.

Check your understanding

5. State the boundary conditions and sequence connection for an LG fault.
Answer: $I_b = I_c = 0$ and $V_a = Z_f I_a$. Thus $I_0 = I_1 = I_2$; the three sequence networks are in series with $3Z_f$. For a solid fault, $V_a = 0$.

Check your understanding

6. Why is zero sequence absent from a pure LL fault?
Answer: No current flows in the unfaulted phase and the two faulted-phase currents are equal and opposite. Their sum is zero, so $I_0 = (I_a + I_b + I_c)/3 = 0$.

Check your understanding

7. Distinguish primary and backup protection.
Answer: Primary protection normally clears faults in the intended protected zone. Backup acts if the primary relay, its trip chain or breaker fails; it may use independent local equipment or delayed remote protection, often with less selectivity.

Check your understanding

8. What makes a protection scheme selective and sensitive?
Answer: Selectivity confines disconnection to the equipment required to isolate the fault. Sensitivity ensures operation for the lowest relevant fault signal while remaining secure under load and permissible transients. Settings must balance both objectives.

Check your understanding

9. Explain induction-relay torque in one equation.
Answer: $T \propto \Phi_1 \Phi_2 \sin\delta$: AC fluxes displaced in space and time induce eddy currents whose interactions produce net torque. For a current-operated arrangement with fluxes proportional to current, operating torque is approximately proportional to I^2.

Check your understanding

10. Give the functional blocks of a static instantaneous OC relay.
Answer: CT/input conversion $\rightarrow$ rectification or conditioning $\rightarrow$ pickup comparator $\rightarrow$ output driver/trip contact. A reference establishes pickup; there is no intentional timing stage, although finite processing/output time remains.

Check your understanding

11. What blocks protect a numerical relay from misleading samples?
Answer: Input isolation/conditioning, an analogue anti-alias filter and appropriate sampling precede digital estimation. The processor checks signals and applies settings/logic. Filtering cannot by itself repair saturated CT information or an incorrect CT ratio/polarity.

Check your understanding

12. State the distance and differential principles.
Answer: Distance estimates impedance from an appropriate voltage/current fault loop and compares it with a reach characteristic. Differential compares compensated currents at a zone's boundaries; substantial imbalance indicates an internal fault, with restraint for measurement error and through-current.

Check your understanding

13. Why is current zero necessary but insufficient for AC interruption?
Answer: The arc receives no instantaneous current input at zero, permitting conductivity to collapse. The contact gap must then recover quickly enough to withstand TRV; otherwise conduction can resume.

Check your understanding

14. Compare the arc media of SF_6 and VCB.
Answer: SF_6 interruption uses gas-flow cooling and rapid insulation recovery, aided by the gas's electron attachment. Vacuum interruption involves contact-metal-vapour plasma; around current zero the vapour/plasma collapses and condenses, allowing rapid gap recovery.

17.3 Numerical drill: questions

N1. On a 100 MVA, 11 kV base, find I_B and Z_B. Express j0.2 p.u. in ohms.

N2. Convert 0.15 p.u. on 25 MVA, 11 kV to 50 MVA, 13.2 kV at the *same physical terminals*.

N3. Find the sequence components of $I_a = 30$, $I_b = 30\angle 120°$, $I_c = 30\angle -120°$ A.

N4. $E = 1$, $Z_1 =$ j0.2, $Z_2 =$ j0.3, $Z_0 =$ j0.1 p.u. Find solid a-ground fault current and sequence currents. Is it larger than the solid three-phase current?

N5. For the same impedances, find solid b-c fault phase currents and sequence voltages.

N6. For the same impedances, find solid b-c-ground sequence currents, both faulted-phase currents, and total earth current. Check I_a.

N7. A CT is 600/5 A. The OC pickup setting is 125% of the 5 A relay rating. At 3000 A primary, find secondary current, primary pickup and pickup multiple. For the standard inverse curve with TMS $= 0.1$, find operating time.

N8. A protected line is $5 +$ j20 Ω primary. Zone 1 is set to 80%. CT ratio is 500/5 and VT ratio is 110000/110. Find the secondary relay reach.

N9. In a compensated differential convention, $I_1 = 8$, $I_2 = 7.2$ A. Use $I_{op} > 0.2 + 0.25 I_{bias}$ with $I_{bias} = (|I_1| + |I_2|)/2$. Does the element operate? Repeat for $I_1 = 8$, $I_2 = 2$ A.

N10. A downstream relay trips in 0.15 s, its breaker clears in 0.06 s, and grading margin is 0.10 s. Find the minimum upstream relay operating time under the specified coordination rule.

17.4 Numerical drill: answers and checks

N1. $I_B = 100/(\sqrt{3}\,11) = 5.248639$ kA; $Z_B = 11^2/100 = 1.21$ Ω; actual reactance is j0.242 Ω/phase.

N2. $0.15(50/25)(11/13.2)^2 = 0.208333$ p.u. Changing the voltage base here is not a transformer-side referral.

N3. This is a pure negative-sequence set: $I_2 = 30$ A, $I_0 = I_1 = 0$. The specified phase order is essential.

N4. $I_0 = I_1 = I_2 = 1/(\text{j}0.6) = -\text{j}1.666667$ p.u.; $I_a = -\text{j}5$ p.u., $I_b = I_c = 0$. Three-phase current is also $1/(\text{j}0.2) = -\text{j}5$ p.u., so magnitudes are equal for these particular data.

N5. $I_1 = -\text{j}2$, $I_2 = +\text{j}2$, $I_0 = 0$ p.u.; $I_b = -3.464102$, $I_c = +3.464102$, $I_a = 0$ p.u. $V_1 = V_2 = 0.6$, $V_0 = 0$, and $V_b = V_c = -0.6$ p.u.

N6. $I_1 = 1/[\text{j}(0.2 + 0.3 \times 0.1/0.4)] = -\text{j}3.636364$; $I_2 = +\text{j}0.909091$; $I_0 = +\text{j}2.727273$ p.u. Then $I_a = 0$, $I_b = -3.936479 + \text{j}4.090909$, $I_c = +3.936479 + \text{j}4.090909$ p.u.; $|I_b| = |I_c| = 5.67727$ p.u. Total earth current = j8.181818 p.u. $V_0 = V_1 = V_2 = 0.272727$, so $V_b = V_c = 0$.

N7. CT ratio = 120; secondary current = 25 A; pickup = 6.25 A secondary = 750 A primary; $M = 4$. $t = 0.1(0.14)/(4^{0.02} - 1) = 0.497976$ s. Breaker interruption time must be added for total clearing time.

N8. Primary reach = $4 + \text{j}16\ \Omega$. Secondary impedance = $Z_{primary}(CTR/VTR)$; $CTR = 100$, $VTR = 1000$, so reach = $0.4 + \text{j}1.6\ \Omega$.

N9. First: $I_{op} = 0.8$ A, $I_{bias} = 7.6$ A, threshold = 2.10 A: no operation. Second: $I_{op} = 6$ A, $I_{bias} = 5$ A, threshold = 1.45 A: operation. These are model decisions, not a substitute for full CT/inrush assessment.

N10. $t_{upstream} \geq 0.15 + 0.06 + 0.10 = 0.31$ s. An additional upstream breaker delay would affect total clearing time, not this stated relay-operating-time inequality.

17.5 Nine-mark practice and answer plans

P1. Per-unit and fault current. Derive the three-phase base impedance and change-of-base formula (3 marks). Convert a generator–transformer–line network to one base and find three-phase fault current (6 marks). Use the integrated example in Section **??** to check your solution. A complete answer shows the local current base and the final kA value, not only p.u. reactance.

P2. Symmetrical components and LG fault. Derive the forward transformation from the three phase equations (3 marks), draw the LG sequence connection and derive its finite-Z_f current formula (3 marks), then solve and verify a numerical case (3 marks). Check all three sequence currents are equal and $V_a = Z_f I_a$.

P3. LL and LLG faults. State phase boundary conditions and derive the LL expression (3 marks). Derive the LLG parallel connection (3 marks). Solve drill N6 and verify unfaulted-phase current and grounded-phase voltages (3 marks). Retaining the signs of I_2 and I_0 is essential.

P4. Protection and relay principles. Explain primary/backup protection and selectivity with a zone diagram (3 marks). Explain directional or distance protection with a labelled characteristic (3 marks). Explain biased differential protection and external-fault stability (3 marks). Use each diagram to connect measurement with the trip decision.

P5. Relay technologies. Explain induction-disc construction and torque (3 marks). Draw static time-OC and instantaneous-OC blocks (3 marks). Draw the microprocessor OC hardware and a pickup/timing/reset flowchart (3 marks). State where auxiliary power, threshold settings and the trip output act.

P6. Circuit-breaker interruption. Explain arc formation and current-zero/recovery conditions (3 marks). Draw and explain an SF_6 puffer breaker (3 marks). Draw and explain a VCB and compare its arc medium with SF_6 (3 marks). Distinguish the breaker interruption process from the preceding relay decision.

17.6 Suggested study sequence

1. Master bases and one generator–transformer conversion before learning the fault formulas. Redo all four per-unit class examples with units visible.
2. Reconstruct the five sequence-component class examples without looking at the matrix. Derive the matrix once; then use it with the stated phase convention.
3. Draw and derive each of the four shunt-fault connections. Redo every class fault/reactor case, including both solid and finite-impedance LG solutions. Use boundary conditions as your answer checker.
4. Learn one functional diagram for each relay principle and technology. Explain what is measured, what is compared, what delays/reset logic apply, and how the trip coil receives the command.
5. Explain each breaker as a time sequence: contacts separate, arc forms, current approaches zero, medium recovers, returning voltage is withstood. Redraw construction and TRV sketches from memory.
6. Use the revision sheets for the final pass, then complete the short questions and numerical drill before checking the answers.

18 References and Further Reading

18.1 Important analytical conventions and nuances

- **LLG sequence current signs:** Negative and zero sequence current signs must be handled consistently for the LLG case. The boundary conditions require $I_a = 0$, $V_b = V_c$ and $I_b + I_c = 3I_0$.
- **Three-phase fault independence:** Fault-type-inactive sequence branches (negative and zero sequence) do not affect the three-phase symmetrical fault result.
- **Per-unit base changes:** Changing voltage base on the same physical side must be clearly distinguished from transforming voltage across a transformer secondary.
- **Protection timing units:** At 50 Hz, one cycle is 20 ms; 0.4 s corresponds to 20 cycles (24 cycles applies only to 60 Hz systems).
- **Directional and differential protection:** Directional operation, zero-sequence transformer paths, induction disc torque, static stage ordering, differential percentage bias and reset behaviour require strict adherence to stated polarity and sign conventions.

18.2 Primary technical references

1. NPTEL, *Power System Analysis*, component transformation and fault analysis: component transformation; fault-analysis introduction; LG fault; LL fault; LLG fault.
2. NPTEL, *Power System Protection*, introductory protection and breaker/relay material.
3. Schweitzer Engineering Laboratories, *Power System Protection* library manual, inverse-time characteristic coefficients; adaptive inverse-time overcurrent technical paper, numerical curve/timing context.
4. Schneider Electric, SF_6 puffer-type interruption explanation, for the gas-flow interruption sequence.
5. ABB, technical review of circuit-breaker interruption, for gas and vacuum interruption context.
6. Siemens, *Vacuum Circuit Breakers*, HG 11.01; 3AF vacuum-breaker manual, for interrupter construction and metal-vapour arc recovery.

18.3 Recommended textbooks

- C. L. Wadhwa, *Electrical Power Systems*, 8th edition, 2023: per-unit analysis, symmetrical components, short-circuit calculations, switchgear/protection.
- V. K. Mehta and Rohit Mehta, *Principles of Power System*, 4th edition, reprint 2020: accessible fault, relay and breaker descriptions.
- Badri Ram and D. N. Viswakarma, *Power System Protection and Switchgear*, 2nd edition, 2011: relay principles, coordination and interruption.
- Hadi Saadat, *Power System Analysis*, 2nd edition, 2002; D. P. Kothari and I. J. Nagrath, *Modern Power System Analysis*, 2nd edition, 2002; B. R. Gupta, *Power System Analysis and Design*, 2006: network and fault-analysis development.